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**Adaptive and Fair methods for Counteracting Negative Influence in
Social Networks**

A project report submitted by: Lahari Mullaguru (6843990)

A project supervised by: Dr. Ehsan Toreini

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University of Surrey
School of Computer Science and Electronic Engineering
Guildford, Surrey GU2 7XH
United Kingdom
Tel: +44 (0)1483 300080

Abstract

The evolving complexity of data-driven applications demands the development of algorithms that not only deliver high performance but also adhere to the principles of fairness. This research presents novel algorithms, to balance both fairness and efficiency in the context of Negative Influence Minimization (NIM) within social networks. Social networks, integral to daily life, facilitate the rapid dissemination of information, which also makes them vulnerable to the spread of misinformation, leading to potentially devastating impacts. The NIM problem seeks to identify a set of nodes in a network that can effectively propagate truth campaigns to counteract misinformation and reduce its overall influence.

To address this challenge, I developed algorithms that integrate fairness constraints with optimization techniques, ensuring that both objectives are met effectively. Truth campaigning, a technique used to combat fake news, involves selecting specific users in the network to disseminate accurate information and counteract the spread of falsehoods. Existing NIM solutions often neglect individual fairness, favoring well-connected users and thereby exacerbating network disparities. This study addresses this issue by focusing on the trade-off between fairness and computational efficiency in NIM algorithms.

Extensive experiments and simulations were conducted to evaluate the performance of these algorithms. Various fairness notions were examined, and algorithms were developed to optimize for individual fairness while maintaining computational efficiency. The methods involve the use of social welfare theory to propose fairness-aware truth campaigning techniques. The performance of the proposed methods is evaluated using both real-world and synthetic datasets, ensuring comprehensive testing across diverse network structures.

The results of my research indicate that the proposed algorithms significantly improve fairness without compromising on efficiency. This demonstrates their potential for broad application in combating misinformation in social networks. The primary contribution of this research is the development of NIM algorithms that balance fairness and computational efficiency. By focusing on optimizing individual fairness and evaluating computational efficiency, the study presents a robust framework for future research and practical applications in designing fair mitigation strategies against misinformation.

In conclusion, the algorithms developed in this research represent a significant advancement in achieving equitable and efficient outcomes in NIM. By ensuring protection for all users, including those with weaker connections, this work provides a thorough and equitable solution to a pressing issue in the digital age. This work lays a foundation for further exploration and refinement of fairness-aware algorithms, with implications for various data-driven domains.

Keywords: Negative Influence Minimization (NIM), Social networks, Fairness-aware algorithms, Individual fairness, Computational efficiency, Truth campaigning, Misinformation mitigation, Group fairness, Social welfare theory

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I certify that work presented in the dissertation is my own unless referenced

Signature: M.Lahari

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Chapter 1. Introduction

1.1 Background

The proliferation of social networks over the past two decades has significantly transformed how information is disseminated and consumed. Platforms such as Facebook, Twitter, Instagram, and YouTube have become integral to daily life, allowing users to connect, share, and access vast amounts of information rapidly. However, this ease of information dissemination has a downside: the spread of misinformation. Misinformation, including fake news, conspiracy theories, and propaganda, can influence public opinion, disrupt social harmony, and lead to real-world consequences such as public health crises and political unrest.

The Negative Influence Minimization (NIM) problem has garnered considerable attention in the fields of network science and data mining. NIM aims to identify a set of nodes within a network to initiate truth campaigns, countering the spread of misinformation and minimizing its overall impact (Budak, Agrawal and El Abbadi, 2011; He and Kempe, 2016). Truth campaigning involves strategically disseminating accurate information to neutralize the effects of misinformation. Despite its importance, existing solutions to the NIM problem often overlook individual fairness, leading to biased outcomes that benefit only a subset of users (Fish, Kun and Lelkes, 2019). Ensuring fairness in NIM is essential to protect all users, especially those who are poorly connected, from the harmful effects of misinformation.

Previous research in NIM and related fields has evolved significantly over the years. Kempe, Kleinberg, and Tardos (2003) laid the foundation for influence maximization in social networks, focusing on selecting influential nodes to maximize the spread of information. Their work introduced the Independent Cascade Model (ICM) and the Linear Threshold Model (LTM), which simulate how information spreads through a network. While their research primarily focused on positive influence maximization, it provided a basis for subsequent studies on NIM.

Budak et al. (2011) extended the ICM to address the spread of misinformation and proposed the Multi-Campaign Independent Cascade Model (MCICM) and Campaign-Oblivious Independent Cascade Model (COICM). These models consider the simultaneous spread of positive and negative information, highlighting the importance of counteracting misinformation with truth campaigns. However, these studies did not address the fairness of the interventions, potentially leading to biased outcomes.

Fish, Kun and Lelkes (2019) introduced the concept of individual fairness in the context of influence maximization, emphasizing the need to consider the equitable treatment of all users in the network. Their work demonstrated that fairness-aware algorithms could mitigate the "rich-get-richer" phenomenon, where well-connected nodes disproportionately benefit from interventions. Inspired by these studies, this research aims to develop algorithms to balance both fairness and efficiency and investigate the trade-off between fairness and computational efficiency in NIM algorithms.

Social networks not only help in connecting people but also serve as crucial platforms for the dissemination of information. These networks have a substantial influence on public opinion and behavior. Unfortunately, they are also susceptible to the rapid spread of misinformation. Misinformation can cause significant harm, including public health crises, political instability, and social discord (He, Qian and Ding, 2022; Lin, Lu and Chen, 2016). The spread of misinformation through social networks necessitates effective strategies to counteract it, making the NIM problem an essential area of study. This research builds on the foundation laid by previous studies (Heidari et al., 2018) to develop fairness-aware algorithms for NIM, addressing the critical need for equitable and efficient solutions to combat misinformation in social networks.

1.2 Research Aim and Objectives

The primary aim of this research is to develop novel algorithms that balance the trade-off between fairness and computational efficiency in addressing the NIM problem. The NIM problem focuses on identifying an optimal set of nodes in a social network to initiate truth campaigns that counteract the spread of misinformation. Misinformation can be disseminated by certain nodes within the network, influencing other nodes and causing the misinformation to spread rapidly (Budak, Agrawal and El Abbadi, 2011; Kempe, Kleinberg and Tardos, 2003). The objective of NIM is to select a set of nodes, known as truth campaign initiators, that can effectively disseminate accurate information to neutralize the misinformation and minimize its overall impact on the network.

Fairness in NIM is crucial because existing solutions often benefit only a subset of well-connected users, ignoring those who are poorly connected (Fish, Kun and Lelkes, 2019). This disparity can exacerbate social inequalities and leave vulnerable populations unprotected from the harmful effects of misinformation. Ensuring fairness in NIM means developing algorithms that consider the well-being of all users, regardless of their network position, and provide equitable protection against misinformation.

The motivation for fair NIM stems from the need to address the limitations of current NIM algorithms that fail to consider individual fairness. These algorithms tend to prioritize nodes with high connectivity, leading to biased outcomes that disproportionately benefit certain users while neglecting others. By incorporating fairness into NIM, I aim to develop solutions that provide equitable protection to all users, ensuring that everyone has an equal opportunity to remain uninfluenced by misinformation.

To achieve these goals, the specific objectives of this research are:

1. To design algorithms that integrate fairness constraints, ensuring equitable protection for all users, including those with poor connections.
2. To evaluate the performance of these algorithms through extensive experiments and simulations on both real-world and synthetic datasets.
3. To conduct a comprehensive trade-off analysis to understand how different factors influence the balance between fairness and efficiency.

By achieving these objectives, this research seeks to contribute to the development of more equitable and efficient strategies for combating misinformation in social networks.

1.3 Research Approach

This research adopts a methodical approach to developing and evaluating fairness-aware NIM algorithms. Initially, various fairness notions are explored to define the criteria for fair influence minimization. Using social welfare theory, algorithms are designed to incorporate these fairness constraints while maintaining computational efficiency. The proposed methods are rigorously tested on diverse network structures using real-world and synthetic datasets. A thorough trade-off analysis examines the interplay between fairness and efficiency, providing insights into optimizing algorithm performance based on specific requirements.

The approach involves several steps:

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1. **Literature Review:** A comprehensive review of existing research on influence propagation, NIM, and fairness in network algorithms. This review provides the necessary background and context for the development of the proposed methods.
2. **Algorithm Development:** Development of fairness-aware NIM algorithms that incorporate social welfare theory. These algorithms are designed to optimize for individual fairness while maintaining computational efficiency.
3. **Experimental Setup:** Design of experiments to evaluate the performance of the proposed algorithms. This includes selecting appropriate datasets, defining evaluation metrics, and setting up the experimental environment.
4. **Performance Evaluation:** Conducting experiments using real-world and synthetic datasets to test the proposed algorithms. The performance of these algorithms is compared with existing fairness-agnostic and group-fairness-based NIM methods.
5. **Analysis of Trade-offs:** Introducing a trade-off parameter for each algorithm and analyzing the balance between fairness and computational efficiency. This analysis provides insights into the effectiveness and efficiency of the proposed methods.

1.4 Dissertation Outline

The structure of this dissertation is organized as follows:

- Chapter 1: Introduction, providing an overview of the background, research aim, objectives, and approach.
- Chapter 2: Literature Review – This chapter reviews existing research on influence propagation, NIM, and fairness in network algorithms, providing the reader with a comprehensive understanding of the current state of the field.
- Chapter 3: Research Approach – This chapter describes the methodology adopted for the experiments, including details of the model parameters, algorithms developed, and how their performance will be assessed.
- Chapter 4: Data Analysis – This chapter presents the results of the experiments, including the performance evaluation of the proposed algorithms and the analysis of the trade-off between fairness and computational efficiency.
- Chapter 5: Discussion – This chapter discusses the implications of the findings, comparing them with existing literature and highlighting the contributions of the research.
- Chapter 6: Conclusion – This chapter summarizes the key findings of the research, discusses the limitations, and suggests directions for future work.

By following this outline, the dissertation provides a comprehensive exploration of the NIM problem, focusing on the development and evaluation of fairness-aware algorithms that balance computational efficiency and fairness in social networks.

Chapter 2. Literature Review

The propagation of influence through social networks has become a critical area of study due to its relevance in various applications, including marketing, public health campaigns, and the spread of information. A significant aspect of this field is the challenge of minimizing negative influence, such as misinformation, while ensuring fairness in the process. This research aims to provide a comprehensive overview of existing research on influence propagation, negative influence minimization (NIM), and fairness in social networks. By understanding the current landscape, I aim to position the research effectively and highlight the contributions of the new algorithms, an adaptive heuristic for balancing fairness and computational efficiency.

2.1 Influence Propagation Problems

Non-Adversarial Influence Maximization

The problem of influence maximization was first introduced in the context of viral marketing, where the objective is to identify a set of key individuals (nodes) in a social network who can maximize the spread of information. Domingos and Richardson (2001) were among the pioneers in this area, introducing the idea of using social networks for marketing purposes by targeting influential individuals. Their work laid the foundation for subsequent studies by demonstrating the potential of social networks for influence propagation.

Kempe, Kleinberg, and Tardos (2003) further formalized the problem and proposed two key models: the Independent Cascade Model (ICM) and the Linear Threshold Model (LTM). These models simulate the spread of influence through a network and have become the standard frameworks for studying influence maximization. In the ICM, each active node has a single chance to activate each of its inactive neighbors, with a certain probability. The process continues in discrete steps until no more activations occur. The LTM, on the other hand, considers the influence exerted by all active neighbors collectively, with each node adopting a new behavior if the total influence exceeds a certain threshold.

Both models have been extensively studied and extended in various directions. For example, Chen et al. (2009) improved the scalability of influence maximization algorithms by developing efficient heuristics that approximate the optimal solution. Their work demonstrated that near-optimal solutions could be achieved with significantly reduced computational effort, making the algorithms more practical for large-scale networks.

Competitive Influence Maximization

As social networks became more complex, researchers extended the influence maximization problem to competitive settings. In these scenarios, multiple campaigns compete for influence within the same network. Bharathi, Kempe, and Salek (2007) introduced one of the earliest models for competitive influence maximization. They proposed strategies for multiple competing campaigns, where each campaign aims to maximize its influence while countering the influence of rivals. Their work highlighted the strategic interactions between different campaigns and the need for robust algorithms to manage these interactions.

Carnes et al. (2007) further explored competitive influence maximization by considering the impact of competitive dynamics on the spread of influence. They developed algorithms that account for the presence of multiple competing campaigns and demonstrated the effectiveness of these algorithms in various network settings. Their work underscored the importance of considering competitive interactions in real-world applications, where multiple entities often vie for influence simultaneously.

More recent studies have continued to refine and expand upon these initial models. For example, He and Kempe (2016) proposed new algorithms that incorporate game-theoretic principles to manage competition among multiple campaigns. Their work provided a deeper understanding of the strategic behavior of campaigns and offered more robust solutions for competitive influence maximization.

2.2. Negative Influence Minimization (NIM)

The spread of misinformation and negative influence in social networks has become a critical concern in recent years. Misinformation can have serious consequences, including undermining public trust, spreading false information, and inciting harmful behaviors. The Negative Influence Minimization (NIM) problem focuses on identifying strategies to minimize the spread of such negative influences by deploying truth campaigns that counteract misinformation.

Early Works and Heuristics

Budak, Agrawal, and El Abbadi (2011) were among the first to formalize the NIM problem. They introduced the Multi-Campaign Independent Cascade Model (MCICM) and the Campaign-Oblivious Independent Cascade Model (COICM) to model the spread of misinformation and truth campaigns simultaneously. In MCICM, the truth and misinformation campaigns propagate independently, each with its own probability of influencing a node. In COICM, both campaigns use the same propagation probability, making it easier to analyze their interactions.

Their work demonstrated that NIM is a complex optimization problem that requires careful consideration of the interactions between competing influences. They proposed heuristic algorithms to identify the optimal set of nodes to initiate truth campaigns, aiming to minimize the overall spread of misinformation. These heuristics, while effective, highlighted the need for more sophisticated methods to handle larger and more complex networks.

Advanced Techniques

Recent advancements in NIM have introduced more sophisticated methods to address the problem. Lin, Lu, and Chen (2016) proposed BIOG, a path-based method for influence blocking. This method focuses on identifying critical paths in the network through which misinformation is likely to spread and strategically blocking these paths to prevent further propagation. Their approach demonstrated significant improvements in minimizing the spread of misinformation compared to earlier heuristics.

He et al. (2022) utilized Q-learning, a reinforcement learning approach, to dynamically adapt strategies for influence minimization. Q-learning allows the algorithm to learn from the environment and improve its decision-making process over time. By continuously updating its strategy based on feedback from the network, the Q-learning approach can effectively counteract misinformation even in dynamic and evolving networks. This method represents a significant advancement in the field, providing a more flexible and adaptive solution to the NIM problem.

These advanced techniques highlight the growing trend of integrating traditional network analysis with modern machine learning approaches. The use of Q-learning and other adaptive algorithms not only enhances the flexibility of NIM strategies but also paves the way for future innovations that can handle increasingly complex and dynamic social networks.

2.3. Fairness in Influence Propagation

2.3.1. Feature-Aware Fairness

Fairness in influence propagation has gained significant attention due to concerns over biases in algorithmic decision-making. Heidari et al. (2018) explored the incorporation of fairness into influence maximization algorithms by balancing efficiency and fairness through social welfare functions. Their work emphasizes the importance of ensuring that influence maximization does not disproportionately favor certain groups, thus preventing biased outcomes.

In the context of influence propagation, adversarial techniques and fairness-aware algorithms, such as those discussed by Gong et al. (2023), focus on equitable influence distribution by addressing biases directly in the model design. These approaches demonstrate that a more equitable spread of influence can be achieved by incorporating fairness metrics, such as the Gini coefficient, into the algorithm's objective functions. This lays the groundwork for developing fairness-aware strategies in influence propagation scenarios.

2.3.2. Group Fairness vs. Individual Fairness

Gong et al. (2023) focused on group fairness, using metrics like the Gini coefficient to ensure fair distribution of influence across different demographic groups. Group fairness aims to provide equitable treatment to predefined groups, ensuring that no single group is disproportionately favored or disadvantaged by the algorithm. Gong et al. developed algorithms that explicitly optimize for group fairness, showing that it is possible to achieve fair outcomes without significantly compromising overall influence spread.

In contrast, Fish et al. (2019) emphasized individual fairness, aiming to treat similar individuals similarly regardless of group membership. Individual fairness focuses on the equitable treatment of individuals based on their network position and influence potential. Fish et al. proposed algorithms that ensure individuals with similar characteristics and positions in the network receive comparable influence opportunities. This approach aligns with the broader goal of fairness in algorithmic decision-making by addressing biases at the individual level.

The distinction between group fairness and individual fairness is crucial for developing comprehensive fairness-aware algorithms. While group fairness ensures equitable treatment of predefined groups, individual fairness addresses disparities at the micro level, providing a more granular approach to fairness. Both approaches are essential for achieving fair influence propagation in diverse and complex social networks.

2.3.3. Social Welfare-Based Fairness

Theoretical Foundations

Social welfare theory provides a robust framework for addressing fairness in influence propagation. Heidari et al. (2018) and Rahmattalabi et al. (2019) applied social welfare functions to measure and optimize fairness in influence propagation problems. These functions aggregate individual utilities to ensure that resource allocation benefits society as a whole, prioritizing the well-being of the most disadvantaged individuals.

Heidari et al. (2018) introduced the concept of social welfare functions in the context of algorithmic fairness, proposing methods to balance efficiency and equity in resource allocation. Their work demonstrated that social welfare functions could be used to design algorithms that achieve fair outcomes while maintaining high

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levels of efficiency. By prioritizing the worst-off individuals in the network, social welfare-based approaches ensure that the benefits of influence propagation are distributed equitably.

Rahmattalabi et al. (2019) extended this framework to influence propagation problems, developing algorithms that optimize social welfare by balancing the trade-offs between fairness and influence spread. Their work showed that it is possible to achieve fair and efficient outcomes by carefully designing algorithms that account for social welfare considerations. These theoretical foundations provide a basis for developing practical algorithms that can be applied to real-world influence propagation scenarios.

Practical Implementations

Practical implementations of social welfare-based fairness in influence maximization and minimization have demonstrated significant improvements. Fish et al. (2019) introduced welfare functions that prioritize the worst-off individuals in the network, aligning with Rawls' theory of justice. Their approach ensures that the benefits of influence propagation are distributed equitably, with a focus on improving the well-being of the most disadvantaged individuals.

Fish et al.'s implementation involved designing algorithms that optimize for social welfare by considering the utility of each individual in the network. By prioritizing individuals with lower initial influence potential, their algorithms achieved more equitable outcomes while maintaining high levels of overall influence spread. This approach demonstrated the practical feasibility of social welfare-based fairness in influence propagation, providing a foundation for future research and applications.

2.3.4. Feature-Blind Individual Fairness

Challenges and Approaches

Achieving fairness without using sensitive attributes, known as feature-blind fairness, poses significant challenges. Kallus et al. (2022) and Borges et al. (2020) addressed these challenges by developing methods that ensure fairness based on structural properties of the network rather than individual attributes. These approaches are crucial for applications where using sensitive attributes is impractical or unethical.

Kallus et al. (2022) proposed methods to achieve fairness by considering the connectivity and influence potential of individuals in the network. Their approach ensures that individuals with similar network positions receive comparable treatment, regardless of their sensitive attributes. This feature-blind approach provides a practical solution to fairness in influence propagation, particularly in scenarios where sensitive attributes are not available or should not be used.

Borges et al. (2020) extended this approach by developing algorithms that optimize for fairness based on network topology. Their methods ensure that influence is distributed equitably by considering the structural characteristics of the network, such as centrality and clustering. These feature-blind approaches provide a promising direction for future research, balancing the need for fairness with practical constraints.

Novel Approaches in NIM

Novel feature-blind approaches in NIM, such as those proposed by Kallus et al., (2022) focus on the structural characteristics of the network to ensure fair treatment of individuals. These methods provide a promising direction for future research, balancing the need for fairness with practical constraints.

For example, Kallus et al. (2022) introduced algorithms that use structural properties like node centrality and shortest path length to ensure that the most influential nodes are targeted for truth campaigns. By focusing on the network structure, these algorithms can achieve fairness without relying on sensitive attributes. This approach is particularly useful in scenarios where sensitive attributes are not available or should not be used due to ethical considerations.

Borges et al. (2020) developed similar algorithms that optimize for fairness based on network topology. Their methods ensure that influence is distributed equitably by considering the structural characteristics of the network, such as centrality and clustering. These feature-blind approaches provide a promising direction for future research, balancing the need for fairness with practical constraints.

2.4. Diffusion Models

Introduction to Diffusion Models

Diffusion models are fundamental in understanding how information, whether true or false, propagates through a network. In the context of NIM, diffusion models simulate how misinformation spreads across social networks, allowing researchers to develop strategies to minimize its impact. The two most commonly used diffusion models are the Independent Cascade Model (ICM) and the Linear Threshold Model (LTM), both of which have been extensively applied to study the spread of influence within networks.

Single Campaign Propagation Models

- **Independent Cascade Model (ICM):** In ICM, the spread of information is modeled as a probabilistic process where each active node has a single chance to activate its neighbors (Kempe, Kleinberg and Tardos, 2003). The likelihood of a neighbor becoming active is determined by the edge weight connecting the two nodes. This model is particularly relevant when simulating the spread of misinformation from an initial set of users (referred to as "seeds") throughout the network.
- **Linear Threshold Model (LTM):** LTM differs from ICM in that it involves a threshold mechanism. Each node in the network has a certain threshold that needs to be met for it to become active. This threshold represents the cumulative influence required from its neighbors to trigger the spread of information (Kempe, Kleinberg and Tardos, 2003).

Two-Campaign Propagation Models

In scenarios where both misinformation and truth campaigns are active, models such as the Multi-Campaign Independent Cascade Model (MCICM) and Campaign-Oblivious Independent Cascade Model (COICM) are utilized. These models extend ICM by allowing nodes to be in one of three states: influenced by misinformation, influenced by truth, or inactive. The competition between these states is central to studying the effectiveness of truth campaigns in counteracting misinformation (Budak, Agrawal and El Abbadi, 2011).

Application of Diffusion Models in NIM

In the NIM problem, diffusion models are used to simulate how misinformation spreads through a network and to identify the most effective nodes for initiating truth campaigns. The objective is to minimize the expected number of users influenced by misinformation by strategically selecting nodes to propagate truth. The choice of diffusion model can significantly impact the outcomes of these interventions, making it essential to select a model that accurately reflects the dynamics of information spread in the given network context.

Authors	Year	Method/Model	Focus	Key Contributions
He et al.	2022	Q-learning-based Approach	Dynamic adaptation of influence minimization strategies	Introduced reinforcement learning (Q-learning) for adaptive NIM strategies
Fish et al.	2019	Greedy and Myopic Heuristics	Fairness in Influence Maximization	Introduced individual fairness in influence maximization using greedy and myopic heuristics
Lin, Lu, and Chen	2016	BIOG (Path-based Influence Blocking)	Blocking critical paths in the network to prevent misinformation spread	Proposed a path-based method that improved the blocking of misinformation pathways
Budak, Agrawal, and El Abbadi	2011	Multi-Campaign Independent Cascade Model (MCICM)	Simultaneous spread of misinformation and truth campaigns	Proposed MCICM to competing campaigns with different propagation probabilities
Bharathi, Kempe, and Salek	2007	Competitive Influence Maximization Models	Influence competition between multiple campaigns	Developed strategies for multiple competing influence campaigns in social networks
Carnes et al.	2007	Competitive Influence Models	Handling competitive dynamics in social networks	Proposed algorithms that account for multiple competing influence campaigns

Table 2.1 Summary of previous Works in Negative Influence Minimization

Chapter 3. Research Approach

This section details the methodologies and algorithmic strategies developed and applied in this research to address the Negative Influence Minimization (NIM) problem, particularly focusing on fairness and efficiency. The approach is grounded in the use of advanced algorithms, including the PROTECT heuristic, Naive PROTECT, and newly developed methods such as Reinforcement Learning for Fair NIM (RL-FNIM), Hybrid Influence and Fairness Maximization (HIFM), Random Forest, and Gradient Boosting algorithms. The design, implementation, and evaluation of these algorithms are thoroughly discussed, with a particular emphasis on their contribution to enhancing fairness in social network interventions.

3.1. Previous Methods

3.1.1. PROTECT Algorithm

The PROTECT heuristic is designed to maximize the protection of a network by strategically selecting a set of seed nodes for truth campaign. The algorithm focuses on identifying and protecting the most vulnerable nodes, ensuring that misinformation is contained as effectively as possible (Unnikrishnan, D. 2023).

Algorithm Design:

- **Initialization:** The algorithm begins by calculating the shortest distance D between all pairs of nodes in the network. The distance from each node to the closest misinformation source is stored in D_M , while the initial distance from nodes to truth nodes is set to infinity in D_T .
- **Vulnerability Identification:** Nodes that are vulnerable (i.e., nodes with a non-infinite distance to a misinformation source) are identified. These nodes are potential targets for protection.
- **Iteration Process:** The algorithm iteratively selects nodes over k iterations:
 - **Save Probability Estimation:** For each iteration, the save probability S_p is estimated using the method described earlier.
 - **Identify Most Vulnerable Nodes:** The nodes with the lowest save probability are identified as the most vulnerable.
 - **Node Selection:** Among candidate nodes, those that can improve the save probability of the most vulnerable nodes are selected. The selection is based on maximizing the number of vulnerable nodes whose save probability is improved.
 - **Tie Breaking:** If multiple nodes have the same impact, additional criteria such as proximity to other vulnerable nodes or overall save count are used to break ties.

Algorithm Steps:

1. Distance Calculation: Compute the shortest distance between all node pairs.
2. Initialization: Initialize distances D_T and identify vulnerable nodes.
3. Iteration: For each iteration:
 - Estimate save probabilities.
 - Identify the most vulnerable nodes.

- Select the node that maximizes the save probability improvement.
- Update distances and candidates for the next iteration.

Implementation and Challenges:

The implementation of PROTECT algorithm is computationally intensive due to the need for multiple distance calculations and Monte Carlo simulations (Unnikrishnan, D. 2023). However, it is highly effective in maximizing network protection, particularly in complex and large-scale networks.

3.1.2. Naive PROTECT Algorithm

The Naive PROTECT algorithm is a streamlined version of the original PROTECT heuristic, designed to provide a faster and more computationally efficient solution to the Negative Influence Minimization (NIM) problem. While it simplifies some aspects of the original algorithm, Naive PROTECT still effectively addresses the need to protect the most vulnerable nodes in a social network from misinformation (Unnikrishnan, D. 2023). The trade-off in using Naive PROTECT is between computational efficiency and the precision of node selection, making it a suitable choice for large-scale networks or when computational resources are constrained.

Algorithm Design:

- **Basic and Maximal Protection Modes:** The algorithm operates in two modes:
 - **Basic Protection:** The algorithm first ensures that every vulnerable node (those closest to misinformation sources) has at least one truth node within a closer distance than any misinformation node. This initial step is crucial to providing a minimum level of protection across the network.
 - **Maximal Protection:** After establishing basic protection, the algorithm proceeds to enhance the protection of these nodes by increasing the number of truth nodes that are closer to each vulnerable node than the nearest misinformation nodes. This step aims to maximize the overall save probability across the network.
- **Simplified Distance Calculations:** Unlike the full PROTECT algorithm, which involves extensive distance calculations and Monte Carlo simulations to estimate the effectiveness of each node selection, Naive PROTECT relies on a more straightforward approach. It uses basic distance comparisons between nodes and prioritizes those with the most straightforward path to improving the network's protection level.
- **Heuristic-Based Node Selection:** The node selection process in Naive PROTECT is driven by a heuristic that quickly evaluates the potential impact of each candidate node. The heuristic prioritizes nodes that can protect the largest number of vulnerable nodes or provide the most significant increase in the minimum save probability.

Implementation:

- **Efficiency:** Naive PROTECT significantly reduces the computational overhead compared to the full PROTECT algorithm. By avoiding complex simulations and relying on heuristic-based decisions, it achieves faster execution times, particularly in large networks.
- **Scalability:** The algorithm is highly scalable, making it suitable for networks with millions of nodes. This scalability is a key advantage in real-world applications where networks are vast and dynamic.

Limitations and Considerations:

- **Approximate Solutions:** While Naive PROTECT is efficient, it provides an approximate solution rather than an exact one. This means that in some cases, more sophisticated algorithms may achieve slightly better protection, but at a significantly higher computational cost (He, Qian and Ding, 2022).
- **Heuristic Dependence:** The effectiveness of Naive PROTECT is heavily dependent on the quality of the heuristic used for node selection. If the heuristic does not accurately represent the network's dynamics, the algorithm's performance may be suboptimal (Unnikrishnan, D. 2023).

3.2. Proposed Methods

3.2.1. Reinforcement Learning for Fair NIM (RL-FNIM)

The RL-FNIM algorithm leverages reinforcement learning (RL) to dynamically learn and adapt strategies for selecting seed nodes in the Negative Influence Minimization (NIM) problem (He, Qian and Ding, 2022). The core idea is to train an RL agent that interacts with the network environment, receiving feedback on the effectiveness of its actions in reducing misinformation spread while maintaining fairness across the network.

Algorithm Design:

- **State Space:** The state of the environment is represented by the current configuration of the network, which includes information on the spread of misinformation, the position of truth nodes, and the remaining budget for additional truth nodes.
- **Action Space:** The agent can choose actions such as selecting a specific node to activate as a truth node. The choice of nodes is guided by the agent's policy, which aims to optimize the overall protection of the network.
- **Reward Function:** The reward is a critical component of the RL-FNIM algorithm. It is designed to balance two objectives: maximizing the reduction in misinformation spread and ensuring fairness in the distribution of influence across the network. The reward can be expressed as:

$$R(s,a) = \text{Fairness_metric}(s') + \text{Influence_reduction}(s')$$

where s' is the new state after taking action a from state s , and the fairness metric could involve calculations such as the Gini coefficient (He, Qian and Ding, 2022; Gong et al., 2023).

- **Learning Process:** The agent learns the optimal policy through Q-learning, where Q-values represent the expected cumulative reward of taking a particular action in a given state. The agent updates its policy iteratively based on the rewards received, adjusting its strategy to improve long-term outcomes.

Implementation:

- **Training:** The RL-FNIM algorithm is trained using a combination of exploration and exploitation. Initially, the agent explores different strategies to gather information about the environment. Over time, it increasingly exploits the strategies that yield the highest rewards.
- **Convergence:** The algorithm continues to update its policy until convergence, where the Q-values stabilize, indicating that the agent has learned an effective strategy for node selection.

- **Computational Requirements:** Implementing RL-FNIM requires significant computational resources, particularly for large networks. However, the dynamic and adaptive nature of the algorithm makes it highly effective in complex and evolving network environments.

Key Advantages:

- **Adaptability:** The RL-FNIM algorithm can dynamically adapt to changes in the network, making it robust in environments where the network structure or the nature of misinformation changes over time.
- **Fairness and Efficiency:** By incorporating fairness directly into the reward function, RL-FNIM ensures that the solution is not only effective in reducing misinformation but also equitable across the network.

3.2.2. Hybrid Influence and Fairness Maximization (HIFM)

The HIFM algorithm combines traditional influence maximization techniques with fairness considerations to create a hybrid approach that optimizes both influence and fairness in the NIM problem. The algorithm seeks to select seed nodes that maximize the spread of truthful information while ensuring that the benefits are distributed equitably across different parts of the network.

Algorithm Design:

- **Fairness-Informed Node Selection:** Each node is evaluated based on a composite score that considers both its potential influence and a fairness metric. The fairness metric could be based on social welfare theory, ensuring that nodes in less connected parts of the network receive appropriate attention.
- **Optimization Objective:** The objective function of HIFM is designed to maximize a weighted sum of influence and fairness, expressed as (Fish, Kun and Lelkes, 2019; Gong et al., 2023; Heidari et al., 2018):

$$\text{Objective} = \alpha \cdot \text{Influence}(\mathbf{T}) + (1 - \alpha) \cdot \text{Fairness}(\mathbf{T})$$

where α is a tunable parameter that balances the trade-off between influence maximization and fairness, and $\mathbf{T}$ represents the set of selected truth nodes. The problem is formulated as a mixed-integer linear program (MILP), where the constraints ensure that the selected nodes contribute to both objectives. The formulation includes variables representing the activation of nodes and the spread of influence, subject to the fairness constraints.

Implementation:

- **Heuristic Solutions:** Heuristics in the HIFM algorithm are crucial for managing the complexity of the MILP formulation in large networks. They provide a way to obtain effective, near-optimal solutions in a fraction of the time required for exact methods. By guiding the selection of truth nodes based on the composite score and iteratively refining this selection, heuristics enable HIFM to balance influence maximization with fairness in a practical and computationally feasible manner.
- **Computational Considerations:** HIFM is computationally intensive, particularly in large networks where the number of possible node combinations is vast. However, the use of heuristics helps to manage this complexity, making the algorithm practical for real-world applications.

Key Advantages:

- **Balanced Approach:** HIFM offers a balanced approach that considers both influence and fairness, making it well-suited for scenarios where equitable distribution of protection is as important as maximizing overall influence.
- **Flexibility:** The algorithm's ability to adjust the trade-off between influence and fairness through the parameter α provides flexibility in tailoring the solution to specific network characteristics and goals.

3.2.3. Random Forest (RF) Algorithm

The Random Forest (RF) algorithm is a machine learning approach that uses an ensemble of decision trees to predict the most influential nodes in a network. Each decision tree in the forest is trained on a different subset of the data, and the final prediction is made by aggregating the predictions of all the trees. The RF algorithm is particularly useful in scenarios where the relationship between node features and influence is complex and non-linear.

Algorithm Design:

- **Training Phase:**
 - **Data Preparation:** The algorithm begins by preparing a dataset of network nodes, where each node is represented by a set of features (e.g., centrality measures, connectivity, node degree).
 - **Feature Selection:** Features are selected that are most likely to contribute to the node's influence within the network. These might include network metrics such as betweenness centrality, closeness centrality, degree centrality, and others.
 - **Building Decision Trees:** The algorithm constructs multiple decision trees during the training phase. Each tree is built from a random subset of the training data and uses a random subset of features at each split, ensuring diversity among the trees.
 - **Ensemble Learning:** The RF algorithm aggregates the results of all decision trees to predict the influence score for each node. The final influence score is determined by averaging the scores from all trees in the forest.
- **Prediction Phase:**
 - **Building the Forest:** Before entering the prediction phase, the Random Forest algorithm has already been trained on the network data. This training involves creating an ensemble of decision trees, each built using a subset of the training data and features (such as node degree, centrality measures, and connectivity patterns). Each decision tree in the forest is trained to predict a specific outcome, in this case, the influence score of each node. Influence score refers to the node's potential to spread information (truthful or misleading) across the network.
 - **Aggregation of Predictions:** Once the forest of decision trees is built, the prediction phase begins. For each node in the network, the algorithm passes the node's features through every tree in the forest.

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- Each tree provides an individual prediction of the node's influence score. These predictions are then aggregated, typically by taking the average or majority vote, depending on the problem setup. The aggregation process smooths out the variability and reduces the risk of overfitting, leading to a more reliable and generalizable influence score.
- **Selection of Seed Nodes:** After all nodes have been evaluated by the forest, the algorithm ranks them based on their predicted influence scores. Nodes with the highest influence scores are identified as the most influential and are selected as seed nodes for the truth campaign. These seed nodes are critical in the NIM problem, as they are the starting points for spreading truthful information to counteract misinformation in the network.
- **Fairness Integration:** To ensure that the selected nodes promote fairness in the network, the algorithm can include fairness metrics in the feature set or adjust the decision-making process to account for disparities in node influence distribution.

Implementation:

- **Scalability:** The RF algorithm is highly scalable and can handle large networks with many features. Its ensemble nature helps to reduce overfitting and improve generalization across different network structures.
- **Computational Considerations:** While the RF algorithm can be computationally demanding, particularly when dealing with a large number of trees, it is parallelizable, which can significantly reduce training time.

Key Advantages:

- **High Accuracy:** RF is known for its accuracy in predicting outcomes, especially in complex, non-linear problems like influence maximization.
- **Robustness:** The ensemble approach reduces the risk of overfitting, making the RF algorithm robust and reliable across different network conditions.
- **Feature Importance:** RF provides insights into the importance of different features in predicting node influence, offering additional understanding of the factors driving influence in the network.

3.2.4. Gradient Boosting Algorithm (GB)

The Gradient Boosting algorithm is another machine learning method that builds an ensemble of decision trees sequentially. Unlike Random Forest, which builds trees independently, Gradient Boosting builds each tree to correct the errors made by the previous trees. This approach is particularly effective for optimizing predictive accuracy and is well-suited to complex influence prediction tasks.

Algorithm Design:

1. Sequential Tree Building:

- **Initial Model:** The process starts with an initial simple model, which could be a constant value that represents the average output in the case of regression or the most frequent class in the case of classification. For influence maximization, this initial model might assign a uniform influence score to all nodes, serving as a baseline.

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- **Model Updates:** Gradient Boosting builds additional models iteratively. In each iteration, the algorithm fits a new decision tree to the residual errors (the difference between the actual values and the predictions made by the ensemble so far). These residuals represent the part of the problem that the current ensemble has not yet learned, allowing the new model to focus on the "hard-to-predict" aspects.
- **Gradient Descent:** The model updates are guided by gradient descent, a method that minimizes the loss function by iteratively moving in the direction that reduces the error. The loss function in Gradient Boosting is a differentiable function that measures the difference between the actual influence score and the predicted influence score.

For influence maximization, the loss function might also incorporate fairness metrics, ensuring that the algorithm not only focuses on reducing error but also on maintaining a fair distribution of influence across the network.

- **Learning Rate:** A crucial hyperparameter in Gradient Boosting is the learning rate, which controls the size of the steps taken in gradient descent. A smaller learning rate results in slower learning but can lead to more accurate models as the algorithm takes smaller steps toward the minimum of the loss function. In influence maximization, adjusting the learning rate helps balance the trade-off between model accuracy and generalization to unseen data.

2. Tree Structure and Regularization:

- **Weak Learners (Decision Trees):** The weak learners in Gradient Boosting are typically decision trees with limited depth. These shallow trees, also known as stumps, are simple models that can capture basic patterns in the data but are not strong predictors on their own. The combination of many such trees, each focusing on correcting the errors of the previous ones, results in a strong, accurate model.
- **Tree Depth and Regularization:** The depth of each tree is controlled by a hyperparameter, often kept small to prevent overfitting. Overfitting occurs when a model learns not only the underlying patterns but also the noise in the training data, leading to poor performance on new data.

Regularization techniques, such as limiting tree depth, reducing the learning rate, and using a maximum number of iterations, help prevent overfitting by ensuring that the model generalizes well to new data.

3. Prediction and Ensemble Aggregation:

- **Final Prediction:** The final model in Gradient Boosting is an ensemble of all the trees built during the training process. The predictions from all trees are aggregated (typically by summing up the contributions from each tree) to produce the final prediction.

For influence maximization, the final prediction would be an influence score for each node in the network, which is then used to rank nodes for selection as seed nodes in the truth campaign.

Implementation:

- **Hyperparameter Tuning:** The performance of the Gradient Boosting algorithm is highly dependent on hyperparameters such as the learning rate, the number of trees, and the maximum depth of each tree. Careful tuning is required to balance the trade-off between model complexity and generalization.

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- **Feature Engineering:** Like Random Forest, Gradient Boosting relies on well-engineered features that capture the relevant aspects of the network structure.

Key Advantages:

- **High Predictive Power:** Gradient Boosting is highly effective at identifying complex patterns in the data, making it a powerful tool for influence prediction.
- **Control Overfitting:** By adjusting the learning rate and other hyperparameters, the algorithm can control overfitting, ensuring robust performance across different datasets.
- **Fairness Consideration:** With appropriate modifications to the loss function, Gradient Boosting can incorporate fairness into its predictions, ensuring that the influence maximization process does not disproportionately benefit certain network segments.

The proposed algorithms—Reinforcement Learning for Fair NIM (RL-FNIM), Hybrid Influence and Fairness Maximization (HIFM), Random Forest (RF), and Gradient Boosting (GB) offer a diverse toolkit for tackling the NIM problem. Each algorithm brings unique strengths, from the dynamic adaptability of RL-FNIM to the high predictive power of machine learning approaches like RF and Gradient Boosting. By integrating fairness considerations into these algorithms, this research ensures that the solutions not only effectively mitigate misinformation but also do so in a way that is equitable across the entire network. These algorithms are designed to be scalable, robust, and adaptable, making them suitable for a wide range of real-world applications in influence maximization and misinformation mitigation.

3.3 Baseline Algorithms

3.3.1. Greedy Heuristic

The Greedy Heuristic algorithm, as proposed by Fish, Kun and Lelkes (2019) is a fundamental method in Fair Influence Maximization that iteratively selects nodes based on their immediate contribution to a predefined influence or fairness metric. At each step, the algorithm identifies the node that provides the maximum incremental gain in this metric and adds it to the set of seed nodes. This process continues until the desired number of seed nodes is selected.

The Greedy Heuristic is widely used due to its simplicity and effectiveness in optimizing short-term outcomes. By focusing on the immediate benefit of each selection, the algorithm can quickly accumulate a significant amount of influence or fairness in the network. However, the primary limitation of this approach is its lack of consideration for the long-term effects of node selection. Because it only considers the next step rather than the overall network structure, the Greedy Heuristic may lead to suboptimal solutions when viewed in the broader context of the network's dynamics. Despite this limitation, it remains a valuable baseline for comparing the performance of more complex algorithms, providing a straightforward measure of how much more advanced techniques can improve upon basic strategies (Unnikrishnan, D. 2023).

3.3.2. Myopic Heuristic

The Myopic Heuristic is another algorithm introduced by Fish, Kun and Lelkes (2019) as part of their work on Fair Influence Maximization. Unlike the Greedy Heuristic, which seeks to maximize immediate gains in influence or fairness, the Myopic Heuristic operates with a more localized focus. It aims to incrementally improve fairness by targeting nodes that are most vulnerable at each step of the selection process.

The strength of the Myopic Heuristic lies in its ability to quickly address imbalances in the network, making it a useful tool in scenarios where immediate, localized improvements are necessary (Unnikrishnan, D. 2023). However, its myopic nature—focusing solely on the next step without considering the cumulative effects—can result in decisions that do not optimize the overall network’s fairness or influence distribution in the long run. As a baseline, the Myopic Heuristic provides a useful contrast to more advanced algorithms, highlighting the importance of a broader, more strategic approach to influence maximization and fairness in complex networks.

3.4 Save Probability Estimation

Save probability estimation is a critical process in determining how likely a node in a network is to remain protected from misinformation after a truth campaign is launched. This estimation is crucial for identifying vulnerable nodes and optimizing seed node selection to maximize the overall protection of the network (He, Qian, and Ding, 2022).

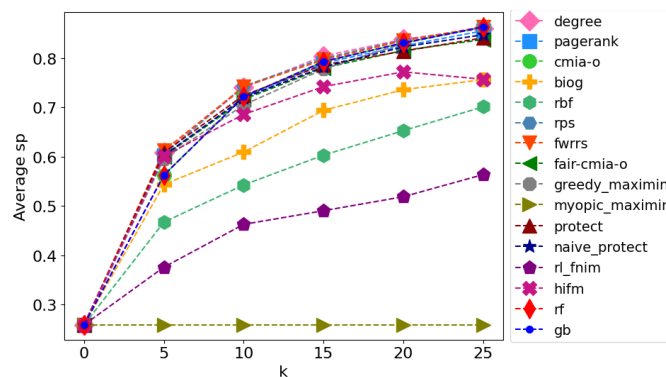
Algorithm Design:

- **Input:** The algorithm begins with inputs that include the graph $G(V, E)$, the set of misinformation sources M , the initial truth set T , the number of simulations R , and influence probabilities α and ϵ .
- **Monte Carlo Simulations:** Multiple Monte Carlo simulations are conducted to simulate the spread of misinformation and truth (Budak, Agrawal, and El Abbadi, 2011). The number of times each node becomes influenced by misinformation is recorded.
- **Save Probability Calculation:** The save probability $sp[i]$ for each node i is calculated as:

$$sp[i] = 1 - \frac{infection_count[i]}{R}$$

where $infection_count[i]$ is the number of times node i was influenced by misinformation across all simulations. This probability indicates how likely a node is to remain uninfected (He, Qian, and Ding, 2022).

Implementation and Complexity: The implementation of save probability estimation involves managing a large number of simulations, especially in large networks. The time complexity depends on the number of nodes, candidate nodes, simulations and accuracy.



(a) Average save probability of email dataset

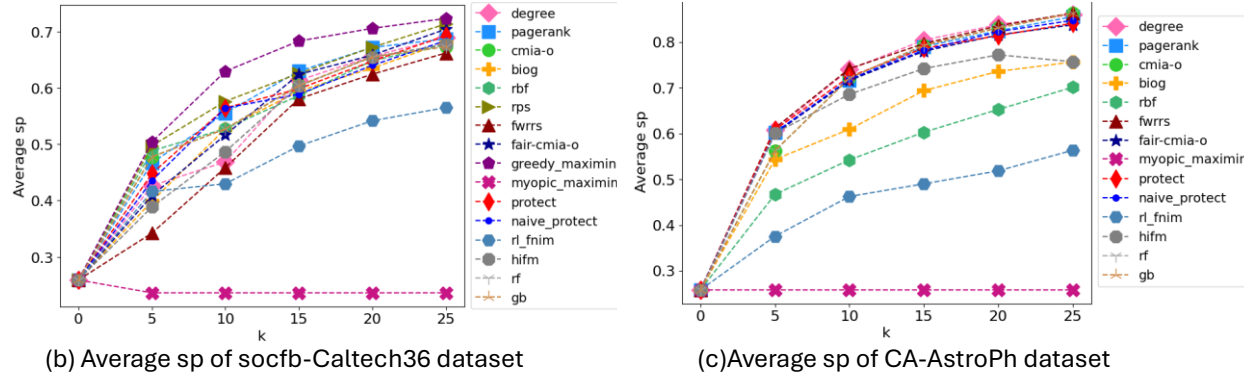


Fig 3.1 Average save probability of Real-world datasets

The graph illustrates the performance of various algorithms in terms of average save probability (sp) across different values of k , which represents the number of truth campaign nodes (or seed nodes) selected.

- The x-axis represents the number of seed nodes k selected for the truth campaign. As k increases, more nodes are used to disseminate truthful information, which ideally should increase the overall protection of the network.
- The y-axis shows the average save probability (sp), which is the likelihood that a node in the network remains uninfluenced by misinformation due to the truth campaign. A higher average sp indicates better performance of the algorithm in protecting the network.
- The observations indicate that specialized algorithms like RL-FNIM, HIFM, and PROTECT are highly effective in mitigating misinformation in social networks, consistently achieving high average save probabilities.

3.5 Fairness-Efficiency Trade-off Analysis

In the context of influence maximization within social networks, achieving a balance between fairness and efficiency is a critical challenge. Fairness refers to the equitable distribution of influence across all nodes, ensuring that no single group or individual disproportionately benefits from or is disadvantaged by the interventions. Efficiency, on the other hand, is concerned with maximizing the overall impact of the influence spread, typically measured by the average save probability, which indicates how effectively misinformation is minimized across the network.

Concept of Fairness-Efficiency Trade-off

The Fairness-Efficiency Trade-off represents the relationship between these two objectives. It captures the idea that improving one of these metrics can sometimes come at the expense of the other. For instance, an algorithm that is highly efficient in terms of influence spread may do so by focusing on the most connected or influential nodes, potentially leading to a less equitable distribution of influence (lower fairness) (Fish, Kun, and Lelkes, 2019; Heidari et al., 2018). Conversely, an algorithm that prioritizes fairness might ensure that influence is more evenly distributed across the network but might not achieve the highest possible efficiency in mitigating misinformation (Gong et al., 2023).

Calculation of the Trade-off

To quantify this trade-off, the performance of various algorithms is assessed based on two key metrics:

1. Fairness (Consistency):

- This metric measures how evenly the influence is distributed across the network. It can be calculated using various fairness measures, such as the Gini coefficient or a consistency metric that evaluates the uniformity of influence across different groups or nodes. A higher consistency value indicates that the influence is more evenly distributed, reflecting greater fairness.

2. Efficiency (Average Save Probability):

- Efficiency is typically measured by the average save probability, which reflects how effectively the algorithm protects the network from misinformation. This is calculated by averaging the probability that each node remains uninfluenced by misinformation due to the truth campaign. A higher average save probability indicates a more effective algorithm in terms of minimizing the spread of misinformation.

Interpretation of the Trade-off

The trade-off between fairness and efficiency is analyzed by comparing how different algorithms perform across these two metrics. Ideally, an algorithm would excel in both fairness and efficiency, but in practice, optimizing for one often involves compromising the other. For example:

- **Efficiency-Focused Algorithms:** These algorithms might prioritize maximizing the overall impact of the truth campaign, selecting the most influential nodes in the network. While this approach can lead to high efficiency, it might neglect less connected or marginalized nodes, resulting in lower fairness.
- **Fairness-Focused Algorithms:** These algorithms aim to distribute influence more equitably across the network, ensuring that all nodes, including those with fewer connections, are adequately protected. While this increases fairness, it might lead to a less efficient spread of the truth campaign, as not all selected nodes may contribute equally to minimizing misinformation.

By analyzing the Fairness-Efficiency Trade-off, this thesis contributes to a deeper understanding of the inherent challenges in influence maximization. It underscores the importance of considering both fairness and efficiency in algorithm design, particularly in applications where both objectives are critical. The trade-off analysis provides insights into how different algorithms balance these competing goals and helps to identify strategies that achieve an optimal balance, depending on the specific needs of the network and the goals of the intervention. This understanding is crucial for developing more effective and equitable methods for mitigating misinformation in complex social networks.

Chapter 4. Data Analysis

4.1 Datasets

In this research, a combination of real-world and synthetic datasets was utilized to rigorously evaluate the proposed algorithms for addressing the Negative Influence Minimization (NIM) problem. By applying the algorithms to diverse datasets, ensuring the robustness and adaptability across different types of networks, characterized by varying structures and sizes.

4.1.1 Real-World Datasets

1. **Arenas Email Dataset:** This dataset reflects an email communication network, where each node represents an individual, and edges denote email exchanges between them. The network's structure often exhibits properties typical of small-world networks, such as short average path lengths and significant clustering. These characteristics make the Arenas Email network particularly suitable for testing the algorithms' ability to manage and mitigate misinformation in environments where information can rapidly circulate within closely connected groups (Leskovec and Kleinberg 2007).

The Arenas Email network is ideal for evaluating the proposed NIM algorithms in a real-world scenario where information flows quickly within tightly connected subgroups. The effectiveness of truth campaigns in such environments depends on the algorithms' ability to identify and target influential nodes that can halt the spread of misinformation within these clusters (Unnikrishnan, D. 2023).

2. **Socfb-Caltech36 Dataset:** The Socfb-Caltech36 dataset is part of the Facebook100 collection, which captures the social networks of various university campuses. Specifically, this dataset maps the social connections of students at the California Institute of Technology (Caltech). Each node in the network represents a student, and each edge represents a friendship between two students. This dataset provides a snapshot of a typical social network on a college campus, where friendships and interactions form a dense and interconnected web of relationships. Such a structure is common in social networks where individuals naturally form close-knit groups based on shared interests or common activities (McAuley and Leskovec, 2012).

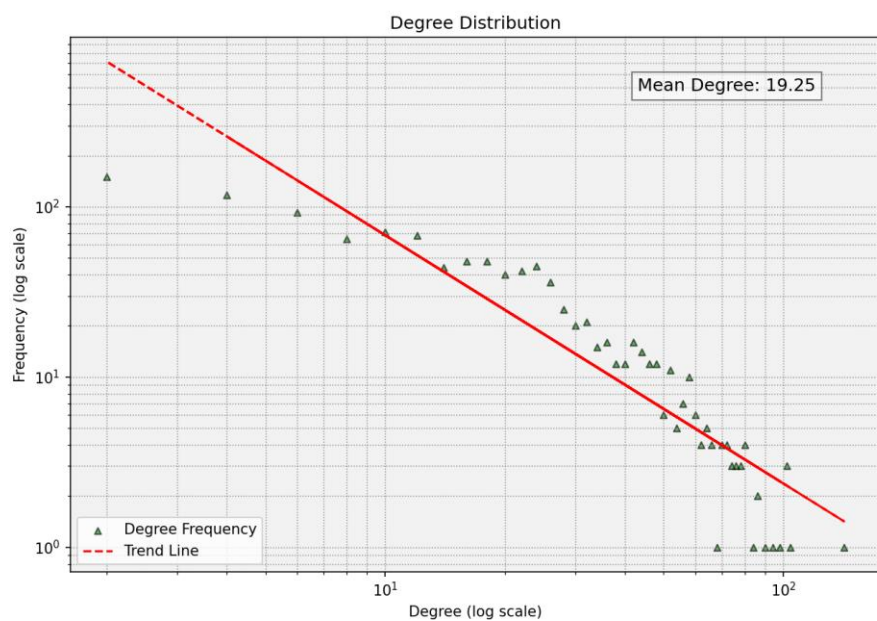
This dataset is particularly useful for testing the proposed algorithms in environments where social influence is strongly shaped by community dynamics. The algorithms must effectively navigate these dense clusters to prevent the spread of misinformation, making this dataset an excellent testbed for evaluating the algorithms' performance in real-world social media contexts (Unnikrishnan, D. 2023).

3. **CA-AstroPh Dataset:** Representing a collaboration network of astrophysicists, the CA-AstroPh dataset consists of nodes that correspond to researchers, and edges that signify co-authorship of papers. The network is characterized by a centralization around key nodes, representing influential researchers, which makes it an ideal case for examining influence propagation. This structure poses a challenge for NIM algorithms, as it involves managing influence in networks where power and connections are concentrated among a few individuals (Leskovec, Kleinberg, and Faloutsos, 2007).

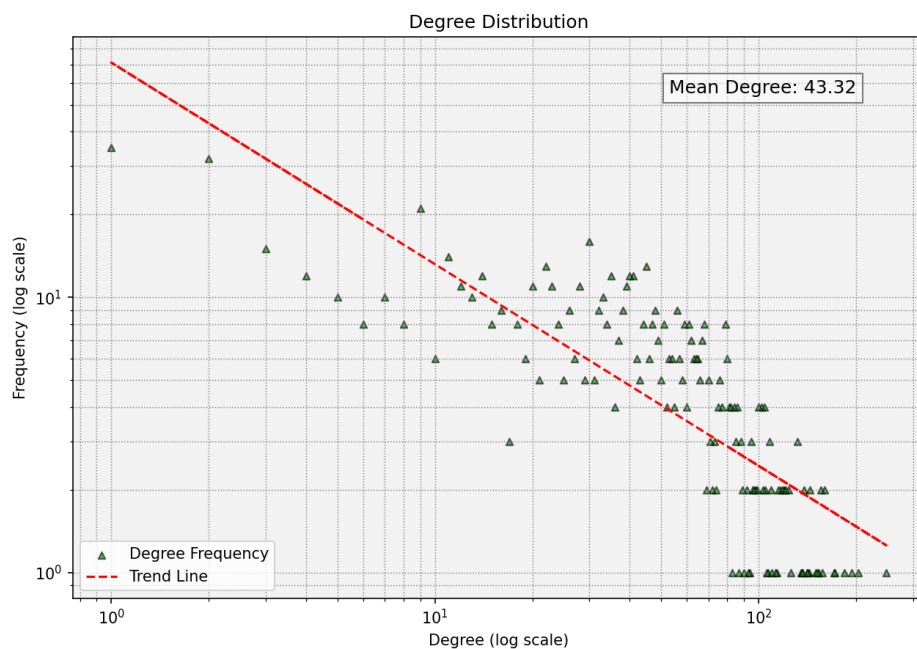
The CA-AstroPh dataset is particularly relevant for studying influence propagation in professional networks where certain individuals (such as leading researchers) have a disproportionate impact on the dissemination of information. The algorithms must be capable of recognizing and leveraging these highly connected individuals to effectively mitigate the spread of misinformation in such a hierarchical and centralized network structure (Unnikrishnan, D. 2023).

Dataset	Nodes	Edges	Average Shortest Path	Clustering Coefficient
Arenas Email	1133	10902	3.6	0.22
Socfb-Caltech36	769	33302	2.3	0.41
CA-AstroPh	18772	396160	4.17	0.63

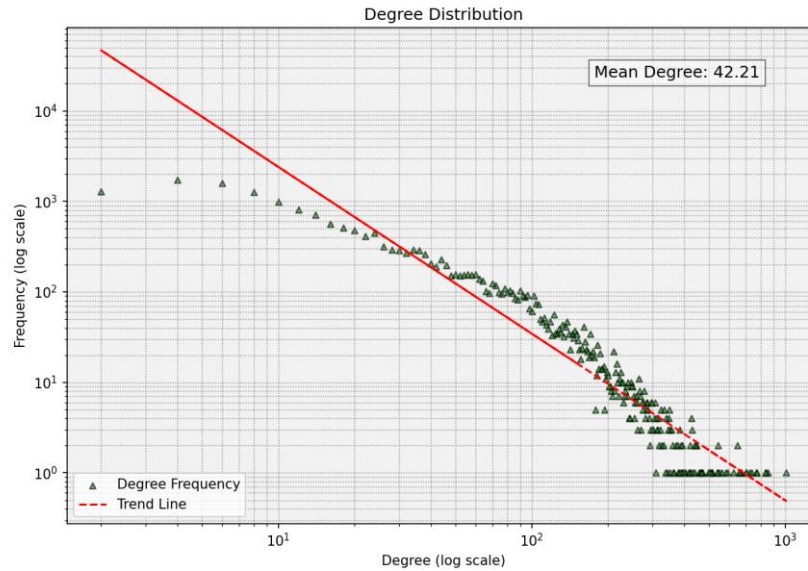
Table 4.1. Characteristics of Real-World datasets



(a) Arenas Email Dataset



(b) Socfb-Caltech36 Dataset



(c) CA-AstroPh Dataset

Fig 4.1 Degree Distribution of Real-world datasets

Arenas Email Dataset: The degree distribution of the Arenas Email dataset is **right-skewed**, a common characteristic in communication networks. This means that while most nodes (representing individuals) have relatively few email connections, there are a few highly connected nodes, or hubs, that play a central role in the network's communication flow. These hubs are crucial in the context of influence propagation, as they have the potential to rapidly disseminate information, including misinformation, across the network.

Socfb-Caltech36 Dataset: In the Socfb-Caltech36 dataset, which represents a social network of students at Caltech, the degree distribution follows a **heavy-tailed pattern**. This indicates that most students have a moderate number of friendships, while a few are exceptionally well-connected, acting as central figures within the network. These highly connected individuals or cliques can exert substantial influence, making them key targets for influence maximization or minimization efforts.

CA-AstroPh Dataset: The CA-AstroPh dataset, which maps collaborations among astrophysics researchers, exhibits a **scale-free degree distribution**. In this network, the majority of researchers have few co-authorships, while a small number of highly prolific authors have a large number of collaborations, positioning them as central hubs within the network. These influential researchers are pivotal in spreading information, making them critical nodes to target in NIM strategies. By focusing on these high-degree nodes, it is possible to effectively manage the spread of misinformation in the academic network, leveraging their centrality to propagate truth campaigns and counter false information.

4.1.2 Synthetic Datasets

HICHBA Algorithm for Synthetic Dataset Generation

The Highly Clustered Barabasi–Albert (HICHBA) algorithm is an advanced method used to generate synthetic networks that possess both scale-free and highly clustered properties. These characteristics are critical for simulating realistic social networks where influence and information flow are heavily influenced by both the degree distribution and the clustering tendencies of nodes (Gutierrez Bierbooms, 2022).

Key Concepts Behind the HICHBA Algorithm:

1. **Scale-Free Networks (Barabasi-Albert Model):** The Barabasi-Albert (BA) model is the foundation of the HICHBA algorithm. It generates networks with a power-law degree distribution, meaning that a few nodes (hubs) have a very high degree (many connections), while the majority have relatively few connections. This mimics real-world social networks where some individuals are much more connected and influential than others (Gutierrez Bierbooms, 2022; Unnikrishnan, D. 2023).
2. **High Clustering:** Real-world networks, particularly social networks, tend to have high clustering coefficients, indicating that individuals within a community are likely to know each other, forming tightly knit groups. The standard BA model does not naturally produce high clustering, so the HICHBA algorithm introduces mechanisms to increase the clustering coefficient.
3. **HICHBA Algorithm Mechanics:** The HICHBA algorithm begins by constructing a small initial network. New nodes are added one by one, similar to the BA model, with the following steps:
 - **Preferential Attachment:** New nodes prefer to attach to existing nodes that already have a high degree, preserving the scale-free nature of the network.
 - **Clustering Control:** Additional connections are added among the neighbors of newly attached nodes to increase the clustering coefficient. This is done selectively, based on the clustering parameters, ensuring that the network remains highly clustered while still reflecting the overall degree distribution typical of a scale-free network.

Based on the HICHBA algorithm, four synthetic datasets were generated to simulate different network structures by varying the clustering coefficients and the degree distribution. Each dataset is designed to test the proposed NIM algorithms under different network conditions, allowing for a comprehensive evaluation of their performance.

1. SN1: This synthetic network consists of 10,000 nodes and 40,000 edges, with a clustering coefficient of 0.9. The high clustering coefficient reflects a tightly-knit network structure, where most nodes are part of highly interconnected communities (Gutierrez Bierbooms, 2022; Unnikrishnan, D. 2023).

- **High Clustering:** With a clustering coefficient of 0.9, this network simulates a social environment where connections are strongly community-based, and most individuals' connections are within their immediate circle of friends or associates.
- **Moderate Density:** The edge count relative to the node count suggests a moderately dense network, where nodes are well-connected but not overly saturated with links.
- This dataset is crucial for testing how well the NIM algorithms perform in environments where misinformation might spread rapidly within close communities. The high clustering could lead to challenges in isolating and controlling misinformation spread within these dense clusters.

2. SN2: SN2 is a larger and denser network with 10,000 nodes and 100,000 edges. It has a high clustering coefficient of 0.9, but the degree distribution exponent is set to 0.1, indicating that there is a higher probability of nodes having similar degrees (Gutierrez Bierbooms, 2022; Unnikrishnan, D. 2023).

- **High Clustering:** Similar to SN1, this network also has a high clustering coefficient, ensuring that the network is composed of tightly-knit communities.

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- **High Density and Uniform Degree Distribution:** The edge count is significantly higher than SN1, leading to a more connected network. The low degree exponent (0.1) suggests that there is less variation in node degrees, with most nodes having a similar number of connections.
- This dataset tests the NIM algorithms' effectiveness in a densely connected environment where influence is more uniformly distributed. The uniform degree distribution can make it harder to identify key influential nodes, posing a challenge for influence maximization strategies.

3. SN3: SN3 consists of 10,000 nodes and 100,000 edges with a lower degree distribution exponent of 0.2 and a high clustering coefficient of 0.9 (Gutierrez Bierbooms, 2022; Unnikrishnan, D. 2023).

- **High Clustering with More Variation in Node Degrees:** While clustering remains high, the lower degree exponent introduces more variation in the degree distribution, meaning there are some highly connected hubs along with many lesser-connected nodes.
- **Scale-Free Nature:** The degree distribution reflects a scale-free network where a few nodes dominate in terms of connections, creating potential central points for influence or misinformation spread.
- SN3 is particularly useful for evaluating how well the algorithms can leverage the presence of highly connected hubs within a clustered network. It simulates scenarios where misinformation could rapidly spread through these hubs unless effectively countered by targeted interventions.

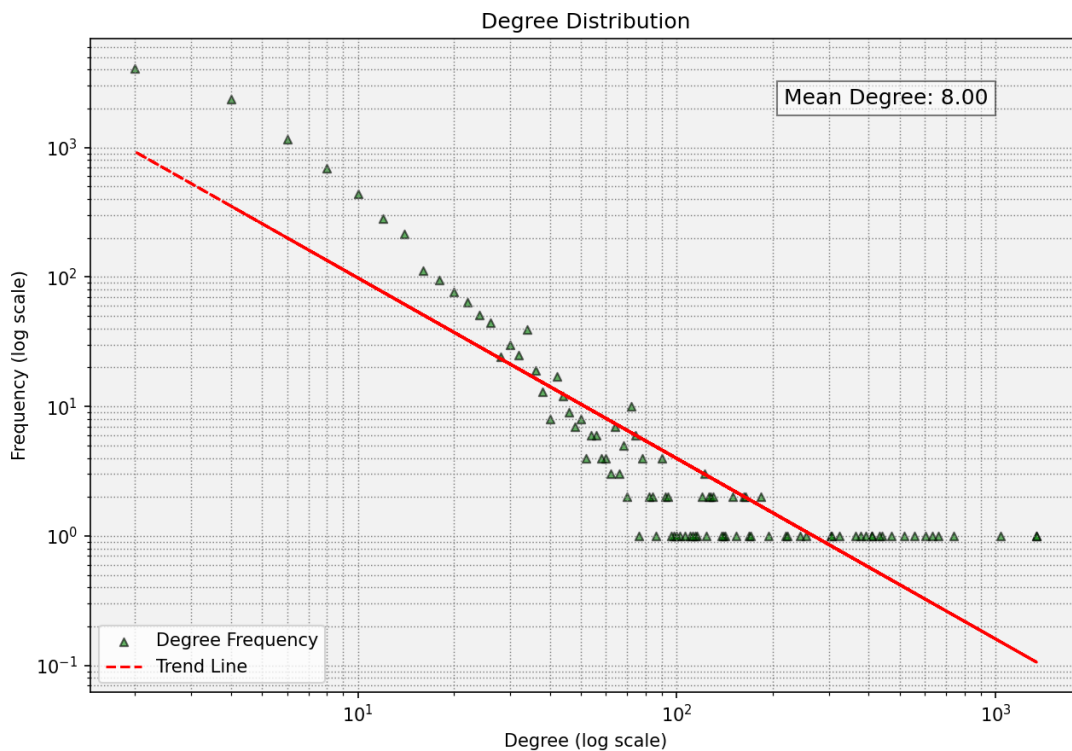
4. SN4: SN4 also has 10,000 nodes and 100,000 edges but features a very low clustering coefficient of 0.2.

- **Low Clustering:** The low clustering coefficient results in a network that is more loosely connected, with fewer tightly-knit communities. This reflects a more dispersed network structure where influence is less localized (Gutierrez Bierbooms, 2022; Unnikrishnan, D. 2023).
- **Wide Degree Distribution:** The low degree exponent ensures significant variability in node connectivity, creating a mix of highly connected hubs and sparsely connected nodes.
- This dataset challenges the algorithms to manage influence in a more dispersed network. The low clustering coefficient means that the spread of influence (or misinformation) may not be as concentrated within clusters, requiring the algorithms to identify and leverage broader network dynamics to achieve effective influence minimization.

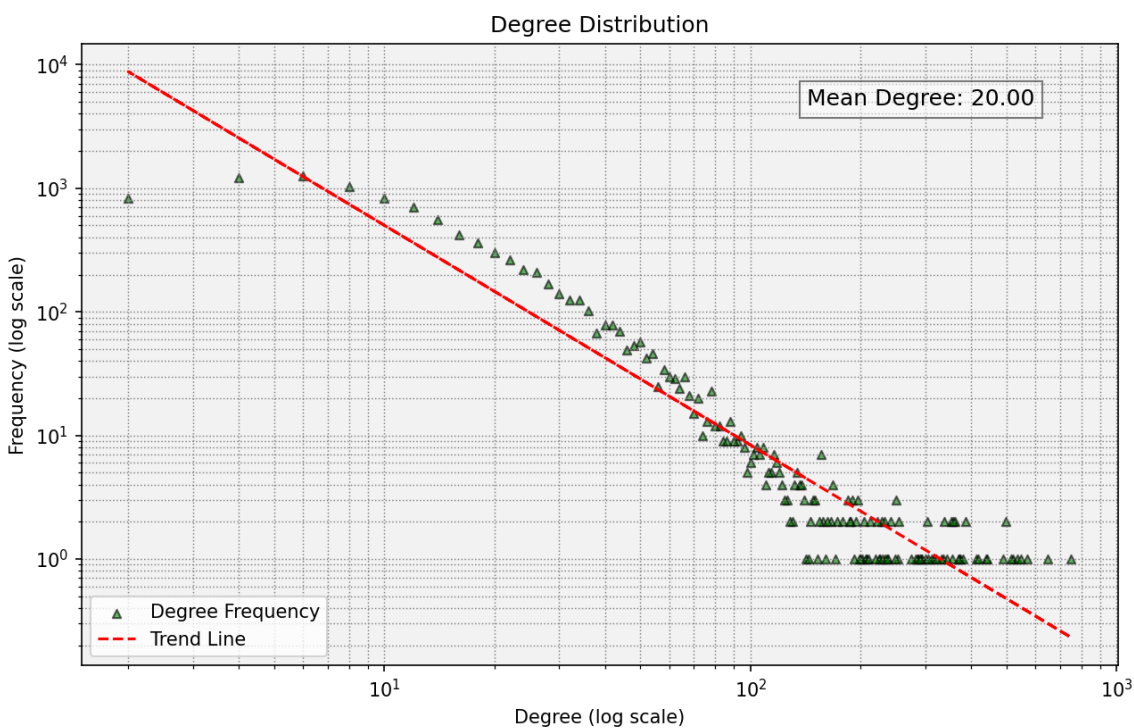
The synthetic datasets generated using the HICHBA algorithm provide a controlled environment for testing the proposed NIM algorithms across different network topologies. By varying the clustering coefficient, degree distribution, and network density, these datasets simulate a range of real-world scenarios, from tightly-knit social communities to dispersed and decentralized networks.

Dataset	Nodes	Edges	Clustering Parameter	Preferential Attachment	Homophily	Clustering Coefficient	Average Shortest Path
SN1	10000	40000	0.9	0.9	0.9	0.31	4.35
SN2	10000	100000	0.9	0.1	0.9	0.48	3.92
SN3	10000	100000	0.2	0.9	0.9	0.13	3.45
SN4	10000	100000	0.9	0.9	0.2	0.40	3.35

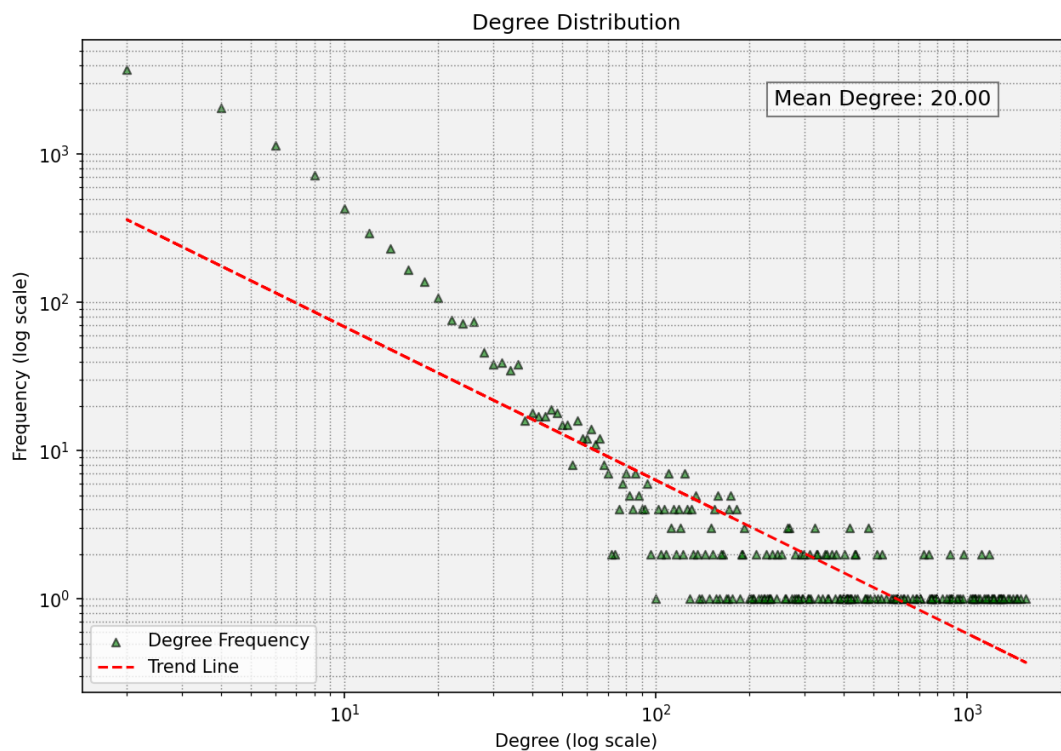
Table 4.2 Characteristics of Synthetic Datasets



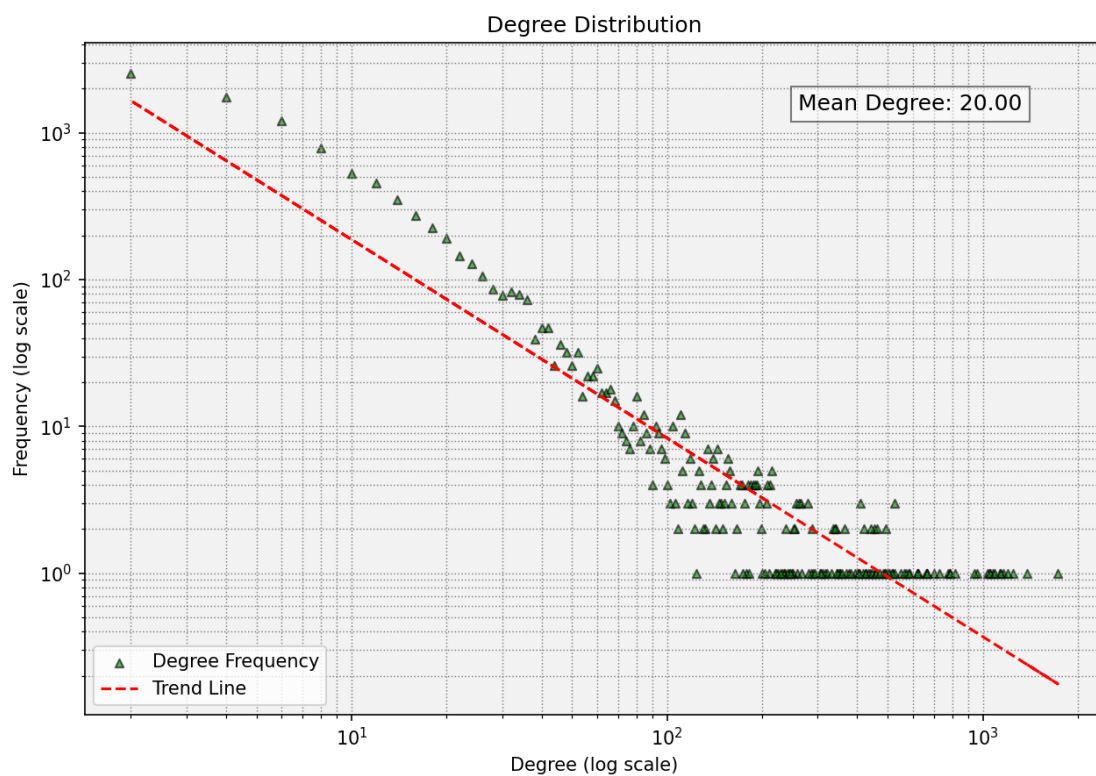
(a) SN1 Dataset



(b) SN2 Dataset



(c) SN3 Dataset



(d) SN4 Dataset

Fig 4.2 Degree Distribution of Synthetic datasets

4.2 Baselines

Proposed Methods

Method	Method Type	Diffusion Models	Baselines Compared
Reinforcement Learning for Fair NIM (RL-FNIM)	Adaptive Heuristic/ML-based	Independent Cascade Model (ICM), Linear Threshold Model (LTM)	Degree Centrality, PageRank, Greedy Heuristic
Hybrid Influence and Fairness Maximization (HIFM)	Heuristic/Optimization-based	ICM, LTM	Degree Centrality, PageRank, Myopic Heuristic
Random Forest (RF)	Machine Learning	ICM, LTM	Degree Centrality, PageRank, BIOG
Gradient Boosting (GB)	Machine Learning	ICM, LTM	Degree Centrality, PageRank

Table 4.3. Proposed Methods and Baselines Compared

Previous Methods

Method	Method Type	Diffusion Models	Baselines Compared
PROTECT	Heuristic/Influence Minimization	ICM, LTM	Degree, Centrality, PageRank, BIOG, Greedy Heuristic
Naive PROTECT	Heuristic/Simplified Approach	ICM, LTM	Degree, Centrality, PageRank, BIOG, Greedy Heuristic
CMIA-O	Path-based	COICM and MCICM	Degree, Proximity, Greedy, Random
BIOG	Path-based	COICM	Greedy, Proximity, Random
TIBM-M	Sampling-based	COICM	TIB-solver, Community based IBM, First Neighbour selection degree, Random
RPS	Sampling-based	MCICM	Greedy
RBF and CDD	Centrality-based	COICM	Betweenness centrality, Degree

Table 4.4. Previous Methods and Baselines Compared

4.2.1 Efficiency-Focused Baselines

Efficiency-Focused Baselines are traditional methods used in the Negative Influence Minimization (NIM) problem, which do not consider fairness when selecting nodes for influence blocking or truth campaigning. These methods focus primarily on efficiency, aiming to minimize the spread of misinformation or maximize the spread of truth without considering how these benefits are distributed among different network nodes or groups (Unnikrishnan, D. 2023).

Common Efficiency-Focused Baselines:

1. **Highest Out-Degree (Degree Centrality):**

- Nodes with the highest number of connections (out-degree) are selected. This method is widely used due to its simplicity and effectiveness in influence maximization and minimization problems (Kempe, Kleinberg, and Tardos, 2003).

2. **PageRank:**

- This method selects nodes with the highest PageRank scores, reflecting their overall importance in the network. PageRank is a common heuristic for various influence propagation tasks (Page et al., 1999).

3. **RBF (Rumor-aware Betweenness Values)** (Kempe, Kleinberg, and Tardos, 2003):

- RBF calculates the number of paths from misinformation sources that pass through each node, selecting nodes with the highest betweenness values for truth campaigning.

4. **BIOG (Blocking Influence Out-Graph):**

- This method identifies critical nodes based on their blocking potential in the Maximum Influence Out-Graph (MIOG) structure, selecting those that can most effectively block misinformation spread (Lin, Lu, and Chen, 2016).

5. **CMIA-O (Community-Aware Influence Minimization Algorithm):**

- A scalable and fast method that uses Maximum Influence Arborescence (MIA) structures to estimate the blocking potential of nodes (Zhang, Chen, and Wang, 2018).

6. **TIB-Solver and TIBM-M:**

- These methods focus on maximizing the block against misinformation spread within a given deadline, using Reverse Reachable sets for sampling (Kallus, Mao, and Zhou, 2022).

4.2.2 Fairness-Focused Baselines

Fairness-Focused Baselines, in contrast, explicitly consider fairness when selecting nodes for influence blocking or truth campaigning. These methods aim to ensure that the benefits of influence propagation or minimization are equitably distributed across different groups or individuals within the network.

Common Fairness-Focused Baselines:

1. **FWRRS (Fairness-aware Weighted Reverse Reachable Sets):**

- Focuses on group fairness by maximizing the minimum save fraction across all detected communities in the network (Zhou and Wu, 2020; Unnikrishnan, D. 2023).

2. **Fair CMIA-O** (Zhang, Chen, and Wang, 2018):

- An adaptation of the CMIA-O method that incorporates fairness by using community detection to ensure that selected nodes provide fair coverage across different groups.

3. **Greedy Welfare Maximin** (Fish, Kun, and Lelkes, 2019):

- This heuristic selects nodes that maximize the minimum save probability, focusing on individual fairness by ensuring that even the most vulnerable nodes are protected.

4. **Myopic Welfare Maximin** (Fish, Kun, and Lelkes, 2019):

- A heuristic that simplifies the greedy approach, aiming to provide a more computationally feasible method for ensuring individual fairness across the network.

These baselines illustrate the difference between methods that prioritize efficiency (fairness-agnostic) and those that integrate fairness considerations (fairness-aware) in the context of influence minimization.

4.3 Evaluation Metrics

Evaluating the performance of algorithms in Negative Influence Minimization (NIM) requires a set of well-defined metrics that measure not only the effectiveness in mitigating misinformation but also the fairness in how this mitigation is distributed across the network. The evaluation metrics used can be categorized into Performance Evaluation Metrics and Fairness Evaluation Metrics, each serving to assess different aspects of algorithm effectiveness.

4.3.1. Performance Evaluation Metrics

These metrics are designed to assess how successfully an algorithm reduces the spread of misinformation or enhances the spread of truthful information within a network. They focus on the algorithm's primary objective which is maximizing the impact of truth campaigns or minimizing the influence of misinformation.

- **Saved Fraction:** The Saved Fraction is a measure of the algorithm's effectiveness in protecting the network's nodes from being influenced by misinformation. It calculates the proportion of vulnerable nodes that are "saved" or protected due to the intervention of the truth campaign.

This metric is computed by comparing the expected number of nodes that would be influenced by misinformation in the absence of any intervention to the number of nodes influenced after the truth campaign has been deployed. The formula is:

$$\text{Saved Fraction is } \frac{\text{Expected affected nodes without truth campaign} - \text{Expected affected nodes with truth campaign}}{\text{Expected affected nodes without truth campaign}}$$

A higher saved fraction indicates that the algorithm is more effective in reducing the spread of misinformation (Unnikrishnan, D. 2023; Chen, Wang, and Yang, 2009).

- **Average Save Probability (sp):** This metric reflects the average likelihood that nodes within the network are protected from misinformation. It is computed by averaging the save probabilities across all nodes in the network. A higher average save probability suggests that, on the whole, the nodes have a greater chance of being protected, indicating a more effective truth campaign.

$$\text{Average Save Probability} = \frac{\text{Total sum of Save Probabilities of all nodes}}{\text{Total number of nodes}}$$

A higher average save probability suggests that the algorithm is effective at protecting the majority of the network's nodes (Lin, Lu, and Chen, 2016; Rahmattalabi et al., 2019; Unnikrishnan, D. 2023).

4.3.2. Fairness Evaluation Metrics

While performance metrics are crucial, it is equally important to ensure that the protection is distributed fairly across the network. Fairness Evaluation Metrics are designed to assess how equitably the influence or protection is distributed among the nodes, particularly focusing on ensuring that even the most vulnerable nodes are adequately protected.

- **Maximin Save Probability:** This metric focuses on the most vulnerable nodes in the network, measuring the minimum save probability among them. It reflects the algorithm's ability to protect even the least connected or most at-risk nodes, thereby ensuring that the benefits of the truth campaign are distributed more evenly (Gong et al., 2023; Heidari et al., 2018).

$$\text{Maximin Save Probability} = \min(\text{Save Probability among vulnerable nodes})$$

A higher Maximin Save Probability indicates that the algorithm provides a baseline level of protection to all nodes, ensuring that no node is left entirely unprotected.

- **Consistency:** Consistency measures how similarly the algorithm treats nodes that are similar in terms of their initial vulnerability to misinformation. High consistency implies that nodes with similar characteristics (such as degree, centrality, or initial exposure to misinformation) receive comparable levels of protection.

Consistency is computed by evaluating the differences in save probabilities between similar nodes:

$$\text{Consistency} = 1 - \frac{\sum_i \sum_j |sp(i) - sp(j)| \times s_{ij}}{\sum_i \sum_j s_{ij}}$$

Here, $sp(i)$ and $sp(j)$ are the save probabilities of nodes i and j , and s_{ij} is a measure of the similarity between these two nodes. A consistency value closer to 1 indicates that the algorithm is fair in distributing protection among similar nodes (Fish, Kun, and Lelkes, 2019; He, Qian, and Ding, 2022).

- **Gini Coefficient:** The Gini Coefficient, traditionally used in economics to measure income inequality, is adapted here to measure inequality in the distribution of save probabilities across the network. It provides a single number that indicates the overall fairness of the algorithm. The Gini Coefficient is calculated as: (Gong et al., 2023; Zhou and Wu, 2020; Unnikrishnan, D. 2023)

$$\text{Gini Coefficient} = \frac{\sum_i \sum_j |sp(i) - sp(j)|}{2 \times |vulnerable| \times \sum_j sp(j)}$$

A lower Gini Coefficient indicates a more equitable distribution of protection, where the save probabilities are more evenly spread across the network's nodes. Conversely, a higher Gini Coefficient suggests greater inequality, with some nodes receiving significantly more protection than others.

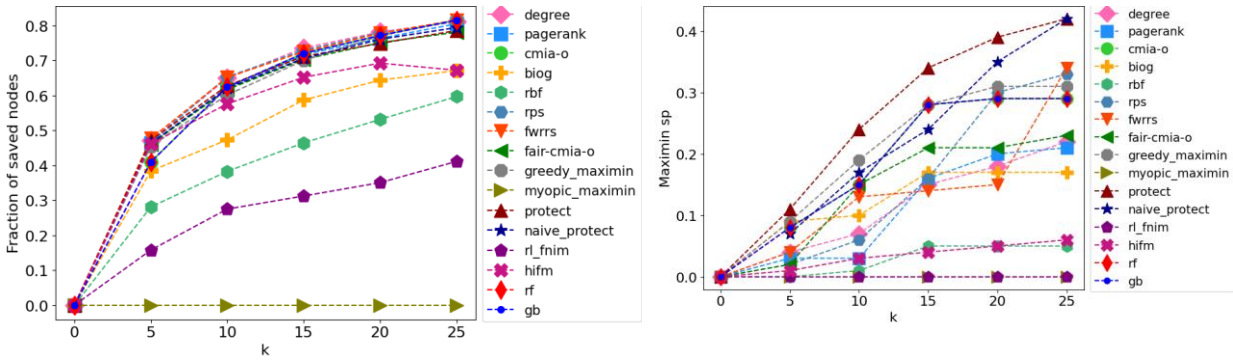
These metrics collectively provide a robust framework for evaluating the effectiveness and fairness of NIM algorithms. They ensure that the methods not only achieve their primary objectives but also distribute their benefits equitably across the network, which is essential in addressing the complex challenges of misinformation in social networks.

Chapter 5. Discussions

The evaluation of the proposed algorithms was conducted using a variety of datasets, including both real-world and synthetic networks. The primary focus of this analysis was to assess the effectiveness of the algorithms in mitigating the spread of misinformation and ensuring fair distribution of influence across the network. The results presented here highlight the performance of the algorithms compared to several baseline methods, with particular attention given to metrics such as Saved Fraction, Average Save Probability, and fairness indicators like Maximin Save Probability and the Gini Coefficient.

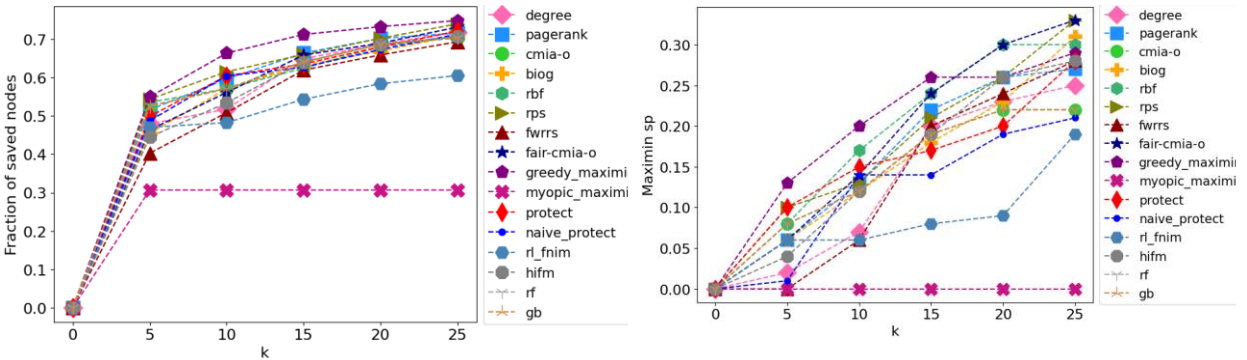
In each experiment, the number of seed nodes k was a critical parameter, representing the number of nodes selected to initiate the truth campaigns. To ensure a thorough evaluation, multiple values of k were tested, reflecting different levels of intervention within the network.

5.1 Fraction of saved nodes and Maximum save probability



(a) Fraction of saved nodes and Maximum sp of Arenas email dataset

The proposed algorithms, particularly **RL_FNIM**, **RF** and **GB**, perform exceptionally well in maximizing the fraction of saved nodes as k increases, these algorithms stand out by providing robust protection to the most vulnerable nodes, especially at higher k values.



(b) Fraction of saved nodes and Maximum sp of socfb-Caltech36 dataset

Adaptive and Fair methods for Counteracting Negative Influence in Social Networks

The proposed algorithms **HIFM**, **RF** and **GB** performing competitively, slightly outperforming traditional methods like **CMIA-O** and **BIOG**. The **RL_FNIM** algorithm, however, show relatively flat performance, indicating it may not be as effective in saving a large fraction of nodes.

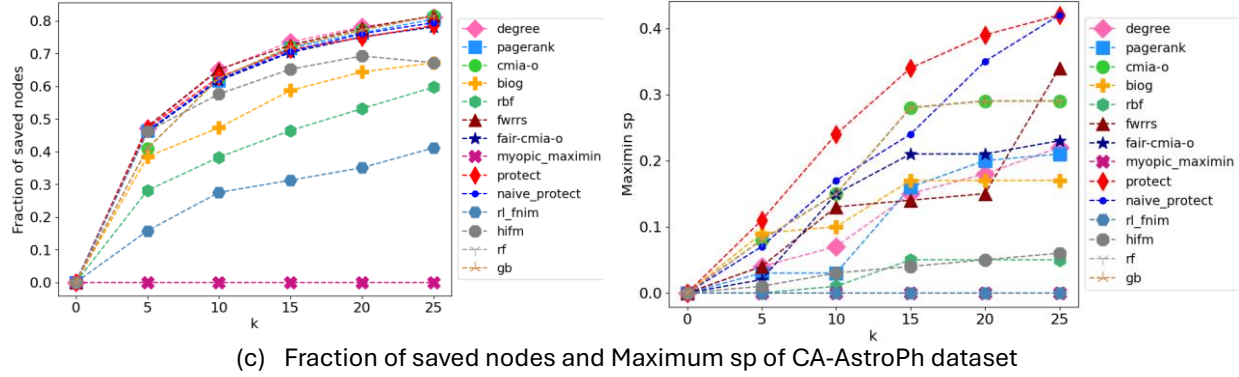
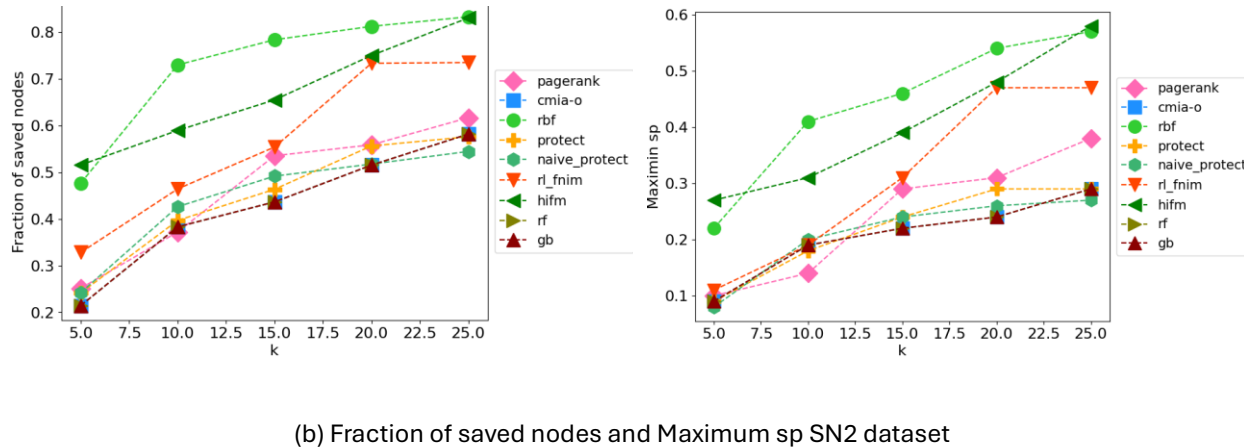
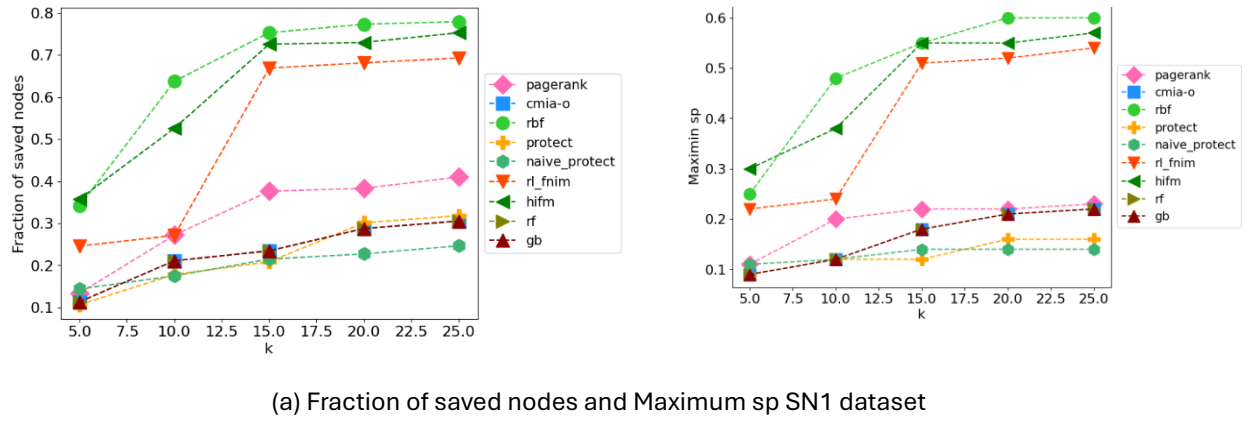
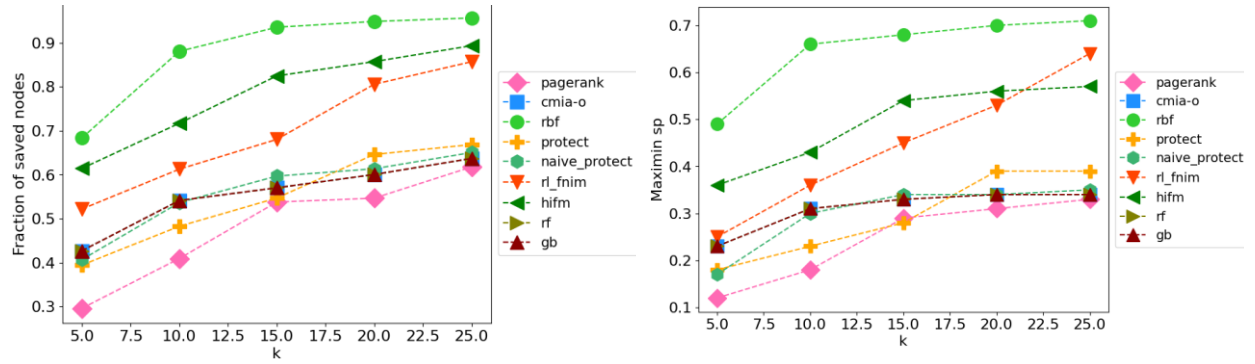


Fig 5.1 Fraction of saved nodes and Maximum save probability of Real-world datasets

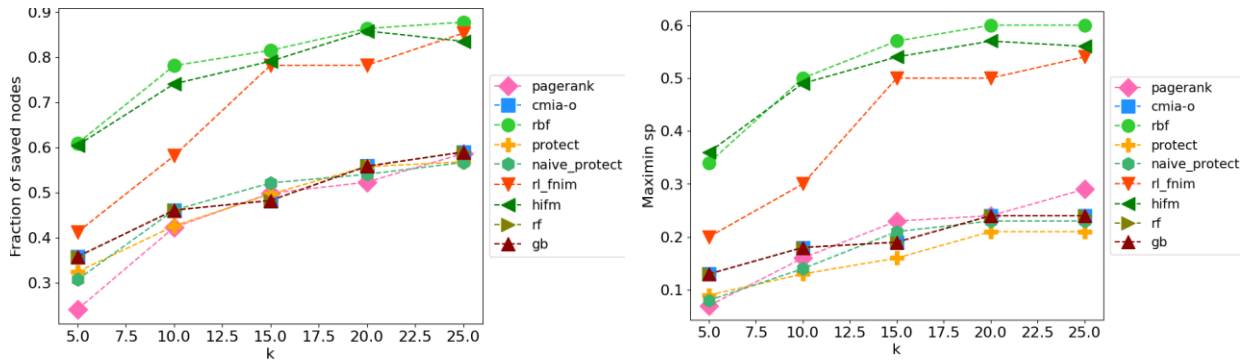
RF and **GB** algorithms perform comparably to traditional methods like **Protect** and **Naïve-protect**, particularly as k increases. The **RL_FNIM** and **HIFM** methods show a flatter curve, reflecting their limited ability to enhance fairness in the network.



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(c) Fraction of saved nodes and Maximum sp SN3 dataset



(d) Fraction of saved nodes and Maximum sp SN4 dataset

Fig 5.2 Fraction of saved nodes and Maximum save probability of Synthetic datasets

As k begins to increase, particularly around $k=5$ and $k=10$, the effectiveness of the algorithms starts to diverge. **RBF** and **protect** quickly rise to the top, showing significant improvements. The proposed algorithms, especially **RL_FNIM** and **HIFM**, also show a noticeable increase in performance, but they may not be as efficient when the number of seed nodes is limited. **Naive protect** shows very modest improvements, highlighting its limited capacity to protect the network effectively.

As k value increases towards the 20 and 25, the effectiveness of all algorithms improves, but **RBF** and **protect** continue to outperform the other algorithms, achieving the highest fraction of saved nodes and Maximin save probabilities. The proposed algorithms **RL_FNIM** and **HIFM**, exhibit substantial improvements, especially at $k=25$.

From $k=0$ to $k=25$, the performance of the algorithms demonstrates that **RBF** and **protect** are consistently the most effective in both saving nodes and ensuring fairness across the network, regardless of the number of seed nodes. The proposed algorithms **RL_FNIM** and **HIFM** performed well, particularly as k increases, indicating their potential for high-resource scenarios. **Naive protect** consistently underperforms across all k values, highlighting the limitations of simpler heuristics in complex network scenarios.

5.2 Average Save Probability and Consistency

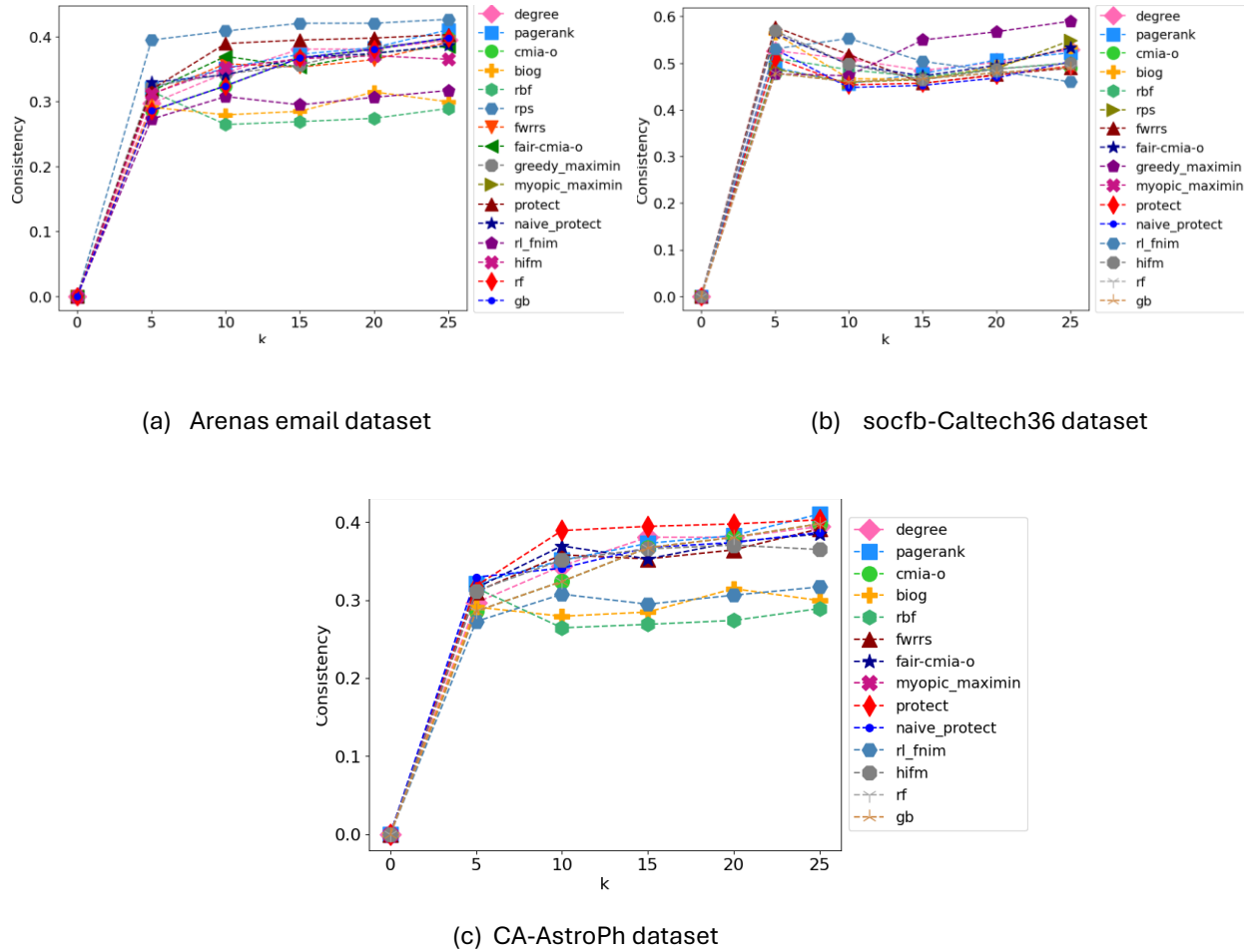


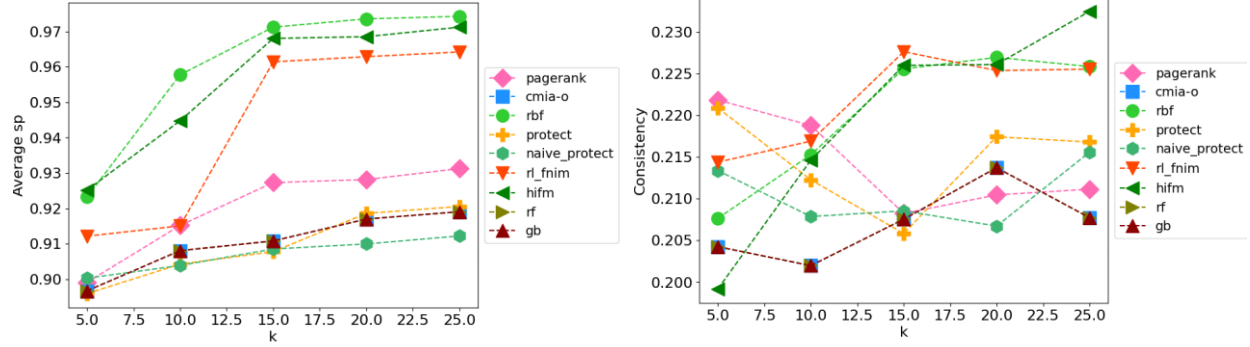
Fig 5.3 Consistency of Real-world datasets

After the initial increase from $k=0$ to $k=5$, consistency tends to stabilize across most algorithms for all three datasets. This suggests that once a certain number of seed nodes is reached, the uniformity of treatment becomes more stable, with only slight variations as k continues to increase up to 25.

Overall, **RF**, **GB**, and **HIFM** are among the most consistent algorithms across all datasets, providing a high degree of fairness in treatment across similarly situated nodes. While other algorithms like **protect** and **RBF** excel in influence minimization, they may require further tuning to achieve similar levels of consistency. The stabilization of consistency after $k=5$ suggests that these algorithms quickly reach a plateau in terms of fairness, which is maintained as more seed nodes are added.

Average probability of real-world datasets discussed in the section 3.4, The proposed algorithms (**RL_FNIM**, **HIFM**, **RF**, and **GB**) show room for improvement in efficiency, particularly in maximizing the save probability. The consistent low performance of **BIOG** indicates that more sophisticated methods are necessary to effectively protect the network from misinformation.

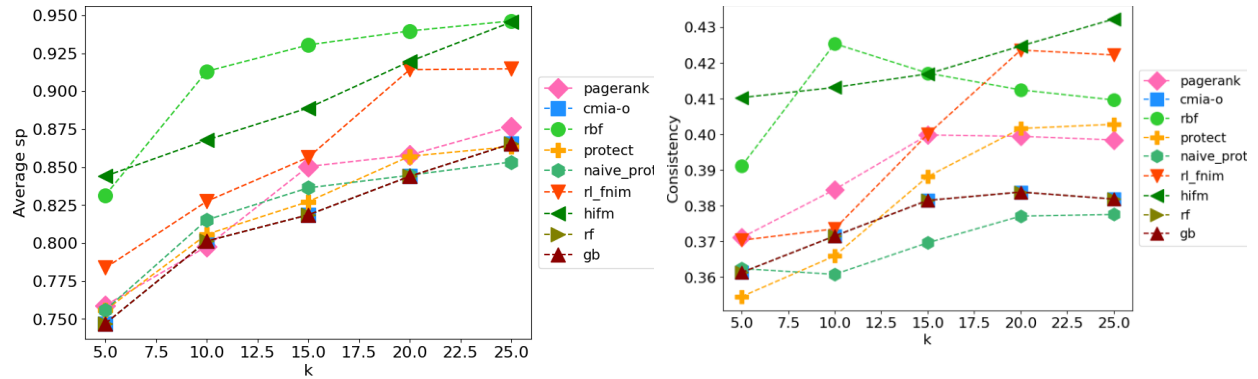
Adaptive and Fair methods for Counteracting Negative Influence in Social Networks



(a) Average sp and Consistency of SN1 dataset

The proposed algorithms, **RL_FNIM** and **HIFM**, along with **RF** and **GB**, perform similarly, with more gradual improvements. They hover around 0.91 to 0.92, indicating moderate effectiveness in saving nodes. **Naive protect** remains consistently lower, suggesting that it is less effective at maximizing the save probability.

The consistency graph reveals more variability across the algorithms, reflecting how uniformly they treat similar nodes in the network. **RL_FNIM** and **HIFM**, exhibit moderate consistency. **RL_FNIM** shows slight fluctuations but generally maintains a stable performance, suggesting that it can balance fairness and efficiency reasonably well.

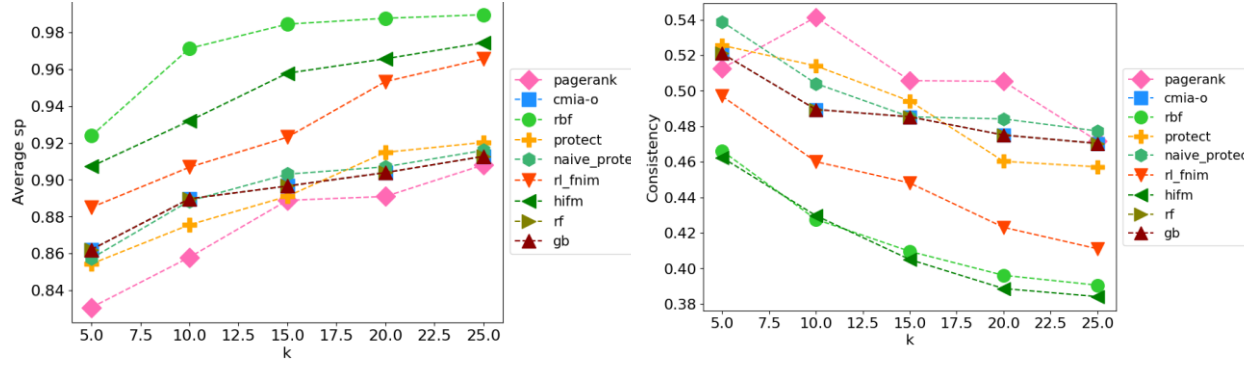


(b) Average sp and Consistency of SN2 dataset

RBF consistently achieve the highest Average Save Probability, peaking above 0.92 as k increases to 25. This indicates that **RBF** is particularly effective in maximizing the number of nodes protected from misinformation. **HIFM** also shows a steady increase, reaching around 0.95 at $k=25$. This underscores its robustness in influence minimization tasks. **PageRank** and **CMIA-O** perform well, with Average Save Probabilities reaching around 0.83 to 0.85 by $k=25$. The proposed algorithms exhibit gradual improvement, and their Average Save Probabilities remain slightly lower, around 0.80 to 0.82.

The proposed algorithm **HIFM** have the higher consistency. **RBF** reaches up to 0.42 at $k=25$. This suggests that **RBF** not only maximizes save probability but also maintains a fairly uniform treatment of nodes across the network. **PageRank** and **CMIA-O** show moderate consistency, with **PageRank** achieving around 0.41 and **cmia-o** around 0.40 by $k=25$. **Naive protect** shows slightly lower consistency, reflecting its simpler heuristic approach. While it does improve with higher k values, its performance remains less consistent compared to more advanced algorithms.

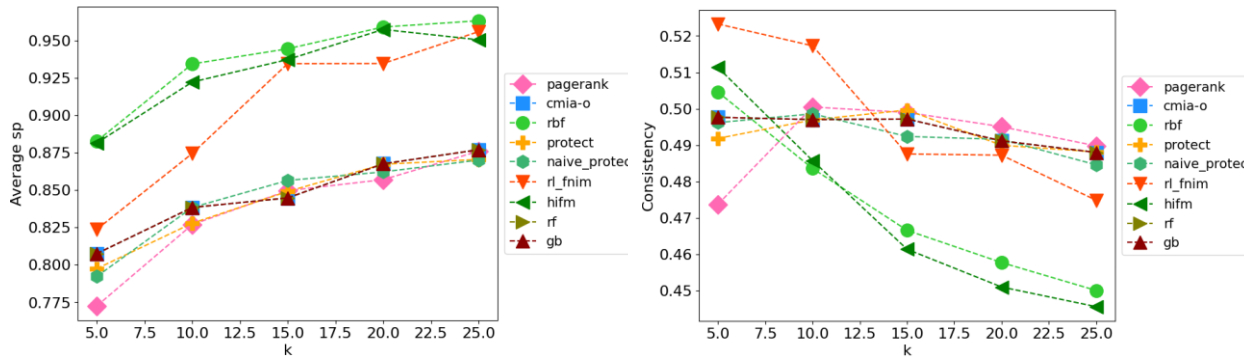
Adaptive and Fair methods for Counteracting Negative Influence in Social Networks



(c) Average sp and Consistency of SN3 dataset

RBF outperform other algorithms, achieving an Average Save Probability close to 0.98 as k approaches 25. This indicates that RBF is highly effective in protecting a large portion of the network from misinformation. The proposed algorithms, including RL_FNIM, HIFM, RF and GB, show gradual improvement, with their Average Save Probabilities ranging between 0.88 and 0.91 at $k=25$.

RF and **GB** shows the highest level of consistency, maintaining a value around 0.50 to 0.52 across the k values. This suggests that these algorithms apply a uniform approach in treating nodes, ensuring fairness in its influence distribution. The RL_FNIM and HIFM show a declining trend in consistency as k increases, with values ranging from 0.44 to 0.38 by $k=25$. This decrease indicates that while these methods can improve overall protection, they may become less fair as more seed nodes are used.



(d) Average sp and Consistency of SN4 dataset

Fig 5.4 Average Save probability and Consistency of Synthetic datasets

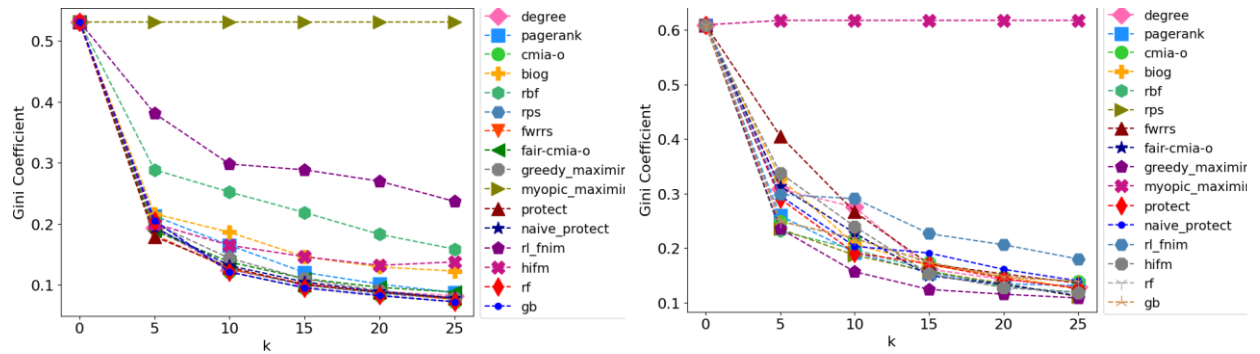
RBF is the most effective algorithm for maximizing the Average Save Probability, but declining consistency suggests a potential trade-off where increased effectiveness comes at the expense of fairness. The proposed algorithms **RL_FNIM** and **HIFM** show gradual increase in average save probability.

PageRank, **CMIA-O** and proposed algorithms **RF** and **GB** maintain the highest consistency across all k values, staying around 0.49 to 0.50. This indicates these algorithms apply a fairly uniform approach in treating similar nodes in the network, which is desirable in scenarios where fairness is as important as efficiency.

5.3 Gini coefficient

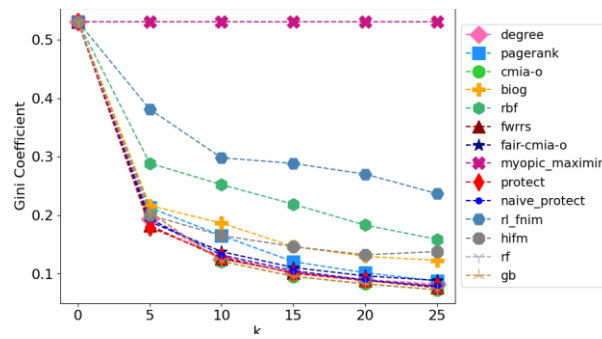
In the context of network analysis and influence propagation, the Gini coefficient can be used to evaluate the fairness of an algorithm by measuring how evenly influence or protection (such as the probability of being saved from misinformation) is distributed across the nodes in a network.

The Gini coefficient ranges from 0 to 1, where 0 represents perfect equality (every node is treated equally or receives the same level of influence or protection), and 1 indicates maximum inequality (a situation where all the influence or protection is concentrated in a single node, leaving others with none).



(a) Arenas email dataset

(b) socfb-Caltech36 dataset

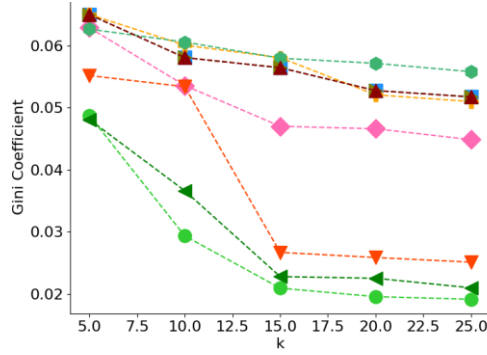


(c) CA-AstroPh dataset

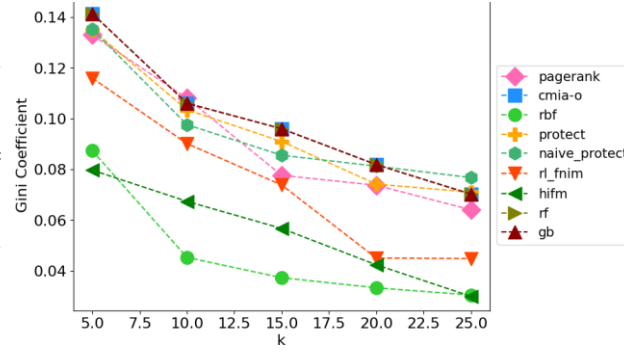
Fig 5.5 Gini coefficient of Real-world datasets

The proposed algorithms, **RL_FNIM**, **HIFM**, **RF**, and **GB**, clearly excel in promoting fairness across the network, as evidenced by their low Gini coefficients. This performance demonstrates their superior capability in distributing influence or protection more evenly among the nodes, minimizing disparities and ensuring that all parts of the network are treated equitably. This makes them highly suitable for applications where both efficiency and fairness are critical, setting them apart from traditional algorithms that may focus more on efficiency at the cost of fairness. Their consistent performance across all values of k underscores their robustness and adaptability to different network conditions, making them valuable tools in the fight against misinformation and other negative influences in social networks.

Adaptive and Fair methods for Counteracting Negative Influence in Social Networks



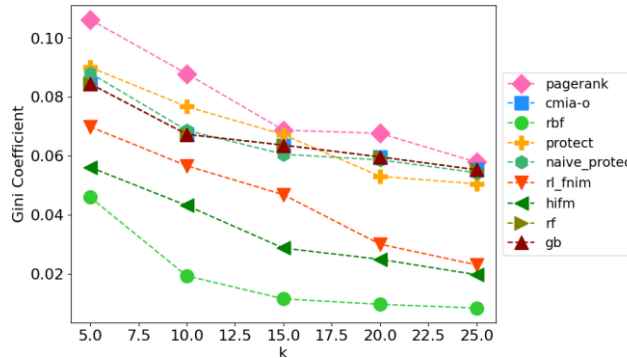
(a) Gini coefficient of SN1 dataset



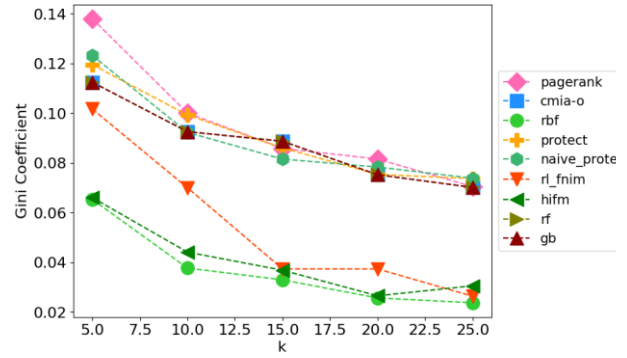
(b) Gini coefficient of SN2 dataset

In both graphs, **HIFM** shows a significant decrease in the Gini coefficient as k increases, indicating that it quickly achieves a fair distribution of influence or protection. By $k=25$, **RL_FNIM** reaches one of the lowest Gini coefficients, highlighting its ability to maintain fairness even as more seed nodes are introduced. **RL_FNIM** also performs exceptionally well, with a consistently low Gini coefficient that decreases further as k increases.

Naive protect and **protect** do not consistently achieve the same low Gini coefficients as **RL_FNIM** and **HIFM**, especially at higher values of k . This highlights the superior ability of **RL_FNIM** and **HIFM** to promote fairness. Traditional algorithms like **PageRank** and **CMIA-O** show higher Gini coefficients, indicating a less uniform distribution of influence. This suggests that while these algorithms may be efficient in other metrics, they may not be as fair in distributing influence or protection evenly across the network.



(c) Gini coefficient of SN3 dataset



(d) Gini coefficient of SN4 dataset

Fig 5.6 Gini coefficient of Synthetic datasets

Across all algorithms, the Gini coefficient generally decreases as k increases. This consistent decrease suggests that as more seed nodes are deployed, the distribution of influence or protection becomes more equitable across the network. **RBF** shows the most significant reduction in the Gini coefficient, reaching near zero levels by $k=25$. This indicates that **RBF** is highly effective in ensuring a very fair distribution of influence or protection across the network.

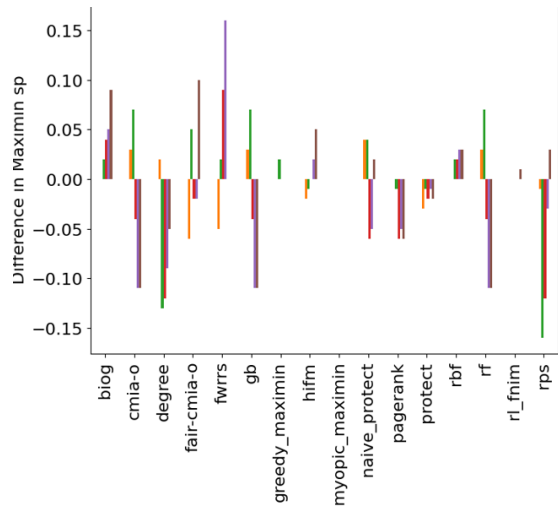
RL_FNIM and **HIFM** also perform exceptionally well, with Gini coefficients consistently decreasing and stabilizing at low levels. These results demonstrate that these algorithms are particularly strong in promoting fairness while distributing influence or protection.

Protect, **CMIA-O**, and **naive protect** maintain a relatively steady performance with moderate Gini coefficients that decrease as k increases but do not reach the low levels. These algorithms provide a fair distribution but are slightly less equitable compared to the top performers. **PageRank** shows the highest Gini coefficients throughout, suggesting it is less effective at achieving equitable distribution. Despite a decrease as k increases, it remains less fair compared to other algorithms, indicating that its influence or protection tends to be concentrated among fewer nodes.

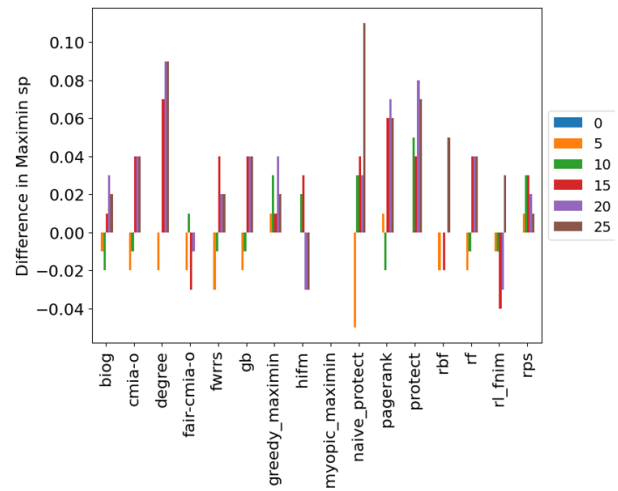
Overall, the proposed algorithms **RL_FNIM** and **HIFM**, stand out as top performers in ensuring fairness, as evidenced by their low and consistently decreasing Gini coefficients. These algorithms demonstrate superior ability to distribute influence or protection equitably across the network, making them excellent choices for scenarios where fairness is a priority. In contrast, traditional algorithms like **PageRank** and **CMIA-O** show higher Gini coefficients, indicating less equitable distribution, which may not be ideal for fairness-sensitive applications.

5.4 Difference in MaxMin Save Probability

The difference in Maxmin Save Probability among algorithms is a critical factor in determining their effectiveness in influence minimization tasks. Algorithms that achieve higher Maxmin Save Probability are more successful at protecting the network, but they might do so at the expense of fairness. Therefore, the choice of algorithm should consider not only the Maximum Save Probability but also the specific requirements of the task, such as the need for fairness and the characteristics of the network.

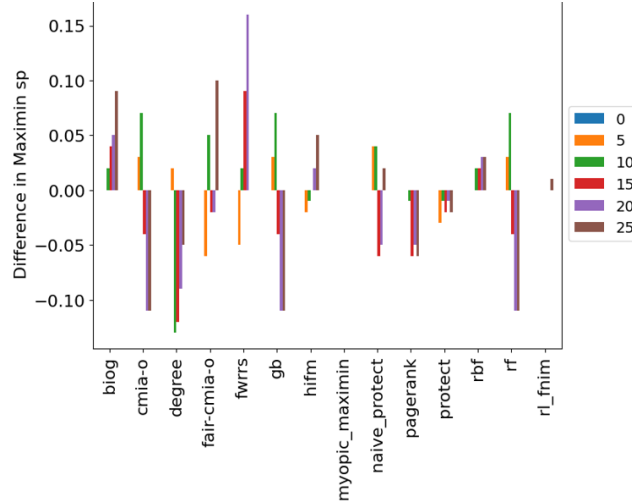


(a) Arenas email dataset



(b) socfb-Caltech36 dataset

Here, positive values indicate an increase in Maxmin Save Probability as k increases, while negative values suggest a decrease.



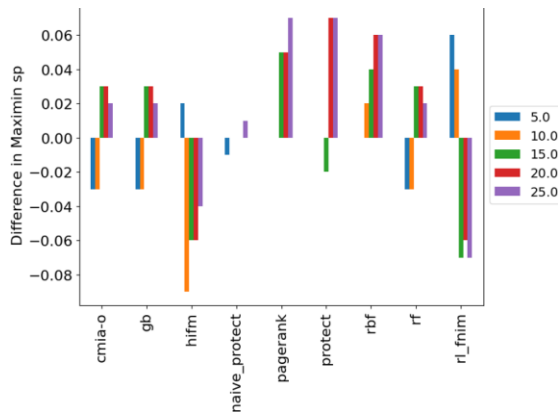
(c) CA-AstroPh dataset

Fig 5.7 Difference of Maxmin save probability in Real-world datasets

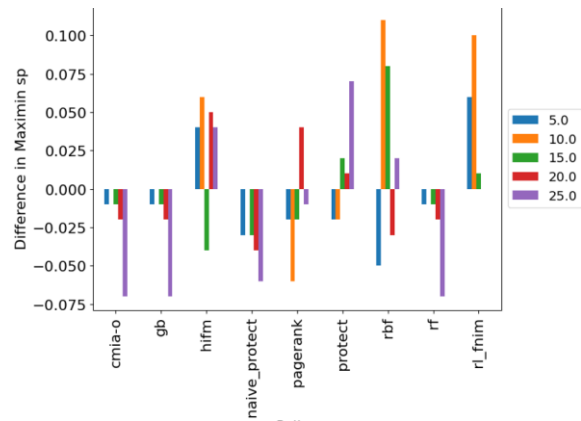
In the email dataset, algorithms like **FWRSS** and **RF** appear to be the most effective in consistently improving fairness across the network as k increases. Meanwhile, the proposed algorithms, **RL_FNIM** and **HIFM**, show potential but also variability in their performance, suggesting that while they are capable of enhancing fairness, they may require further refinement to perform consistently across different network scenarios.

In the socfb-Caltech36 dataset, **RL_FNIM** and **HIFM** consistently improve the Maximum Save Probability as the number of seed nodes increases. Algorithms like **PageRank** and **CMIA-O** may require additional strategies to better leverage increased resources for improving fairness, as indicated by their consistently negative differences in Maximin Save Probability.

In CA-AstroPh dataset, **RL_FNIM** shows a slight positive difference at lower k values but tends towards neutral or slightly negative differences as k increases. **PageRank** and **protect** algorithms show negative differences in Maximin Save Probability, particularly at higher k values. **BIOG** and **degree** display a mix of positive and negative differences, with no significant trends indicating consistent improvement.



(a) SN1 dataset



(b) SN2 dataset

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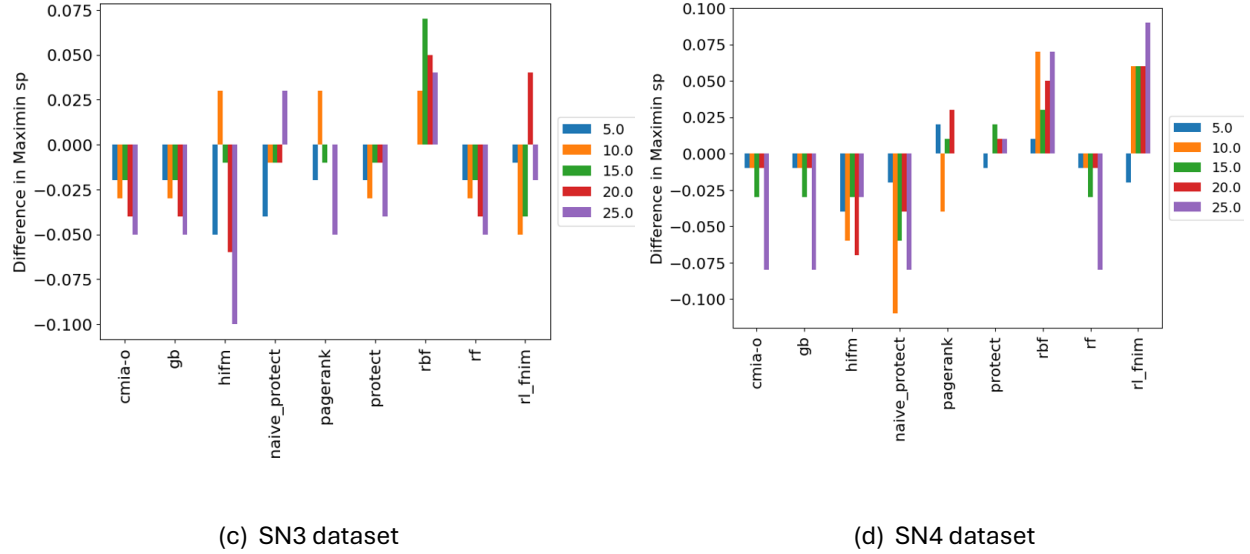


Fig 5.8 Difference of Maxmin save probability in Synthetic datasets

In SN1 dataset, **RL_FNIM** demonstrates positive differences, particularly at $k=25$, indicating its potential to enhance fairness. **CMIA-O** and **RF** consistently show negative differences, suggesting that these algorithms struggle to improve fairness in SN1 as more seed nodes are introduced. **RBF** and **protect** show consistent positive differences, especially at higher k values.

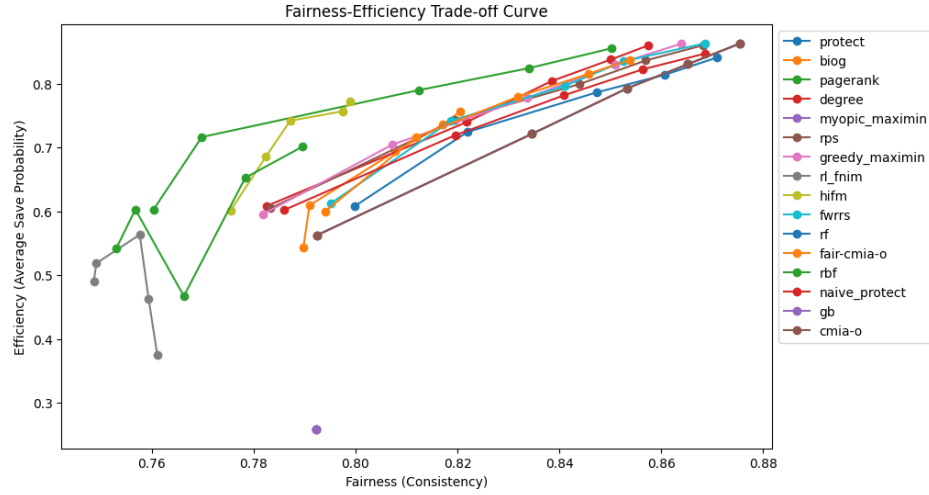
In SN2, SN3 and SN4 datasets, **RBF** and **RL_FNIM** show strong positive differences at higher k values, particularly at $k=10$ and $k=25$. This indicates these algorithms are effective in enhancing fairness in these networks by improving the minimum save probability. **GB** and **HIFM** show a mix of positive and negative differences, indicating inconsistent performance across different k values. **CMIA-O** and **RF** display predominantly negative differences, suggesting that these algorithms may not be well-suited for improving fairness in these networks.

5.5 Trade-off Analysis

Trade-off analysis is a critical aspect of evaluating the performance of algorithms, particularly when balancing conflicting objectives such as efficiency and fairness. In the context of influence minimization or maximization, and especially when dealing with social networks, trade-off analysis helps in understanding how well an algorithm performs in terms of spreading influence (or protection) efficiently while ensuring that the process is fair and equitable across the network.

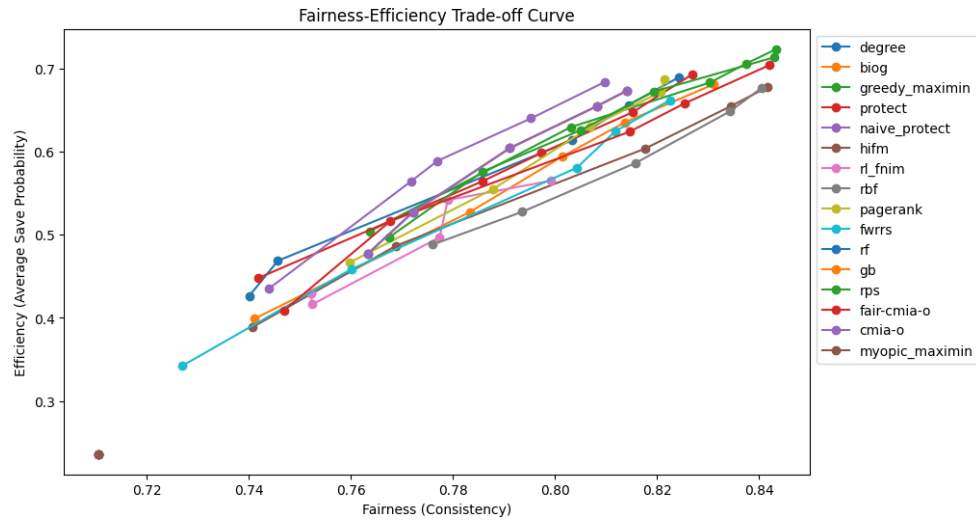
Trade-Off Curves are often plotted to show the relationship between efficiency (Average Save Probability) and fairness (Consistency). Efficiency typically refers to the algorithm's ability to maximize the spread of influence or protection across the network using the seed nodes. In this context, it often means achieving a high overall save probability or minimizing negative influence effectively. Fairness is concerned with how evenly the influence or protection is distributed across the network. It ensures that all nodes, regardless of their position or connectivity, have a fair chance of being protected or influenced. Fairness metrics include the Maxmin Save Probability, which focuses on the least protected nodes. An algorithm that maximizes the Maxmin Save Probability ensures that even the least protected node has a significant chance of being protected, which is crucial for fairness. However, focusing on maximizing this metric might lead to a reduction in overall efficiency, as the algorithm might need to allocate more resources to less connected or less influential nodes.

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(a) Arenas email dataset

In the Arenas email dataset, the proposed algorithms **RL_FNIM**, **RF** and **GB** demonstrate strong performance in balancing the trade-off between fairness and efficiency. They are positioned favorably on the trade-off curve, indicating that they not only maximize the overall save probability but also ensure that this protection is distributed equitably across the network. This makes them highly effective in scenarios where both fairness and efficiency are critical considerations. In contrast, while some traditional algorithms like **protect** and **PageRank** achieve high efficiency, suggesting that these algorithms might concentrate protection on more influential nodes, leaving others less protected.



(b) socfb-Caltech36 dataset

In the socfb-Caltech36 dataset, **RL_FNIM** algorithm is positioned toward the upper right of the trade-off curve, indicating that it successfully achieves both high fairness and high efficiency. It consistently offers a strong balance, ensuring that protection is not only maximized but also equitably distributed across the network. **HIFM** also performs well, showing competitive results in both fairness and efficiency. **RF** and **GB** also show competitive results, making them valuable options when both fairness and efficiency are crucial. While

traditional algorithms like **protect** and **BIOG** excel in efficiency, they might not distribute protection as fairly, making the proposed algorithms particularly well-suited for tasks where equitable treatment across the network is essential.

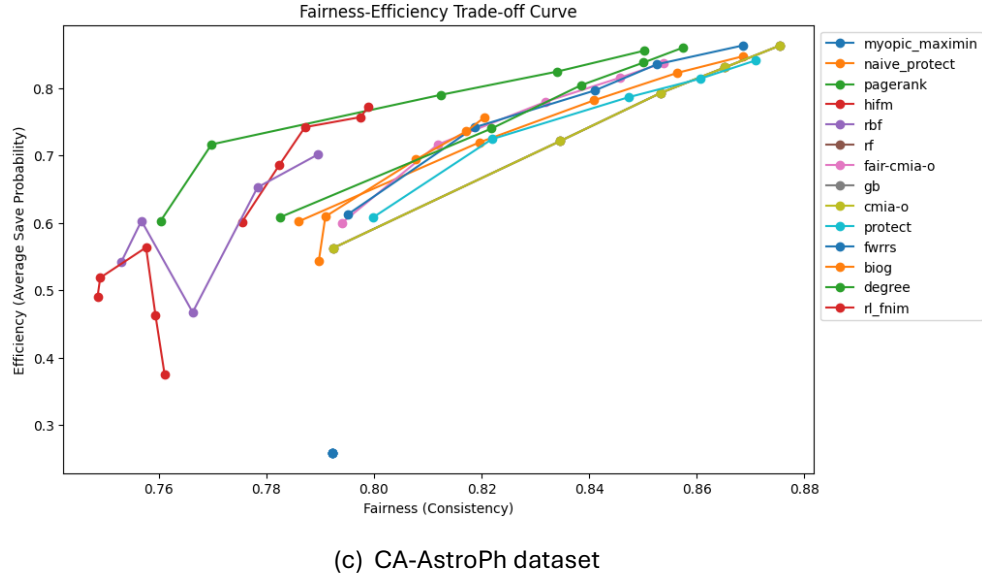


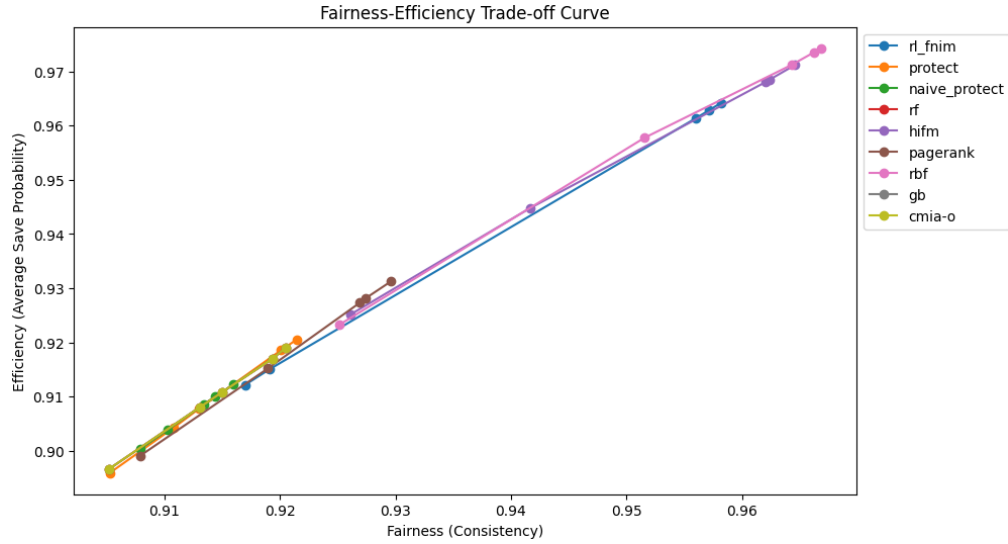
Fig 5.9 Trade-off analysis of Real-world datasets

In the CA-AstroPh dataset, the proposed algorithms **RL_FNIM**, **HIFM**, **RF**, and **GB** excel in balancing the trade-off between fairness and efficiency. **RL_FNIM** and **GB** are particularly strong, showing that they can maintain both high overall protection and equitable distribution across the network. **RF** and **HIFM** also perform well, indicating that they are effective choices when both fairness and efficiency are critical. While traditional algorithms like **RBF** and **Degree** excel in efficiency, they may not achieve the same level of fairness as the proposed algorithms. **CMIA-O** and **Fair-CMIA-O** tend to perform lower on the trade-off curve, indicating that they may struggle to achieve high levels of both fairness and efficiency in the CA-AstroPh dataset. **PageRank** and **protect** algorithms achieve good efficiency but at the cost of lower fairness, indicating that they might favor certain nodes over others, leading to a less equitable distribution of protection.

In all the three datasets, there is a consistent positive correlation between fairness and efficiency, indicating that improving fairness generally leads to better overall efficiency (measured as average save probability). However, the extent to which this relationship holds, and the relative performance of the algorithms vary between datasets, highlighting the importance of considering the specific characteristics of each network.

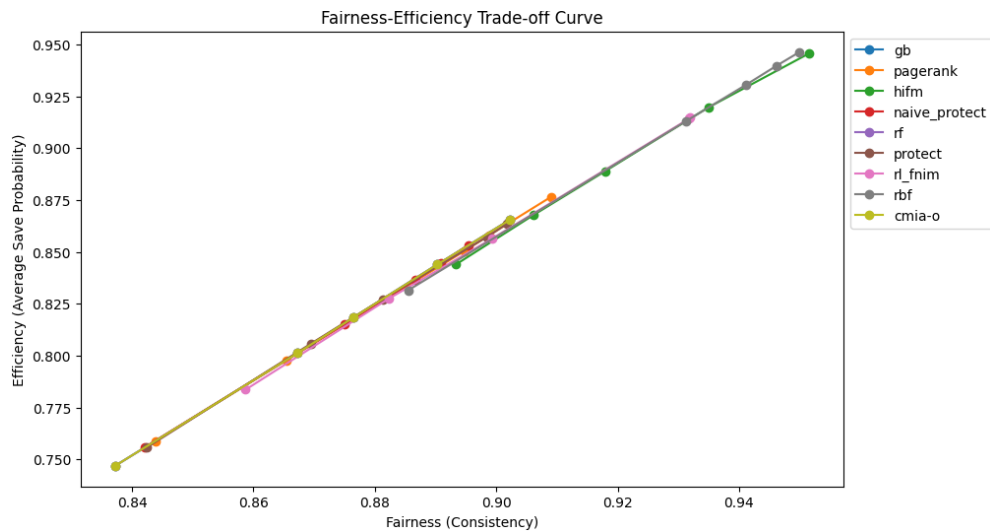
Across all real-world datasets, **RL_FNIM** and **HIFM** are the standout performers, consistently achieving a strong balance between fairness and efficiency. **RF** and **GB** also perform well, making them reliable choices, particularly in scenarios where efficiency is slightly prioritized. The traditional algorithms, while effective in specific contexts, generally do not achieve the same level of fairness as the proposed algorithms, highlighting the advantages of the newly developed methods in balancing these critical objectives across diverse social network datasets.

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(a) SN1 dataset

The SN1 dataset shows a very tight clustering of algorithms along the trade-off curve, indicating that most of the algorithms perform similarly in balancing fairness and efficiency. The slight variations among them are crucial in understanding their specific strengths. **RL_FNIM** demonstrates its ability to maximize protection while distributing it equitably across the network. **HIFM** is also closely clustered near the top of the trade-off curve, indicating that it maintains high fairness and efficiency. **RF** is a reliable choice for scenarios requiring consistent performance. **GB** aligns closely with **RF**, indicating that it too provides a good balance between fairness and efficiency. The traditional algorithms like **protect** and **PageRank** perform well but may not match the optimized balance provided by the proposed methods.

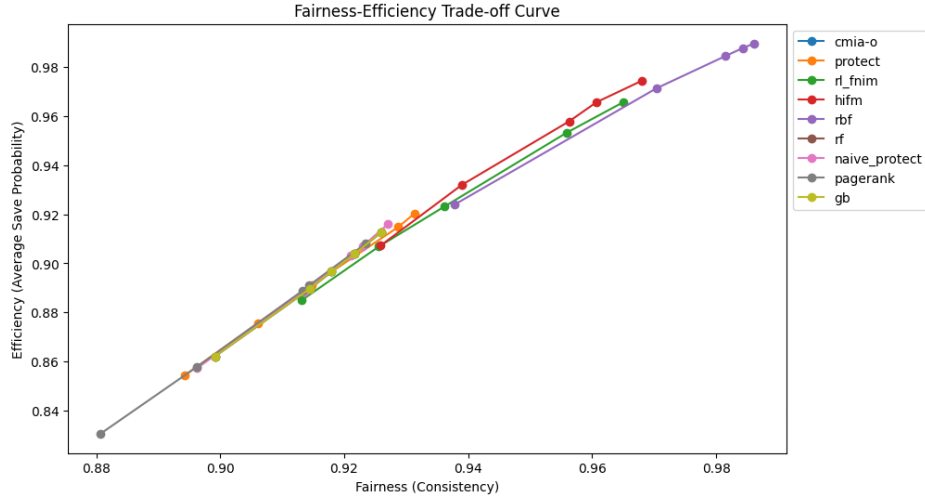


(b) SN2 dataset

Similar to the SN1 dataset, the SN2 dataset shows a tight clustering of algorithms along the trade-off curve. **RL_FNIM** continues to demonstrate robustness across different synthetic datasets. **Protect** performs similarly to the proposed algorithms, especially **RL_FNIM** and **HIFM**, showing that it is an effective baseline in synthetic

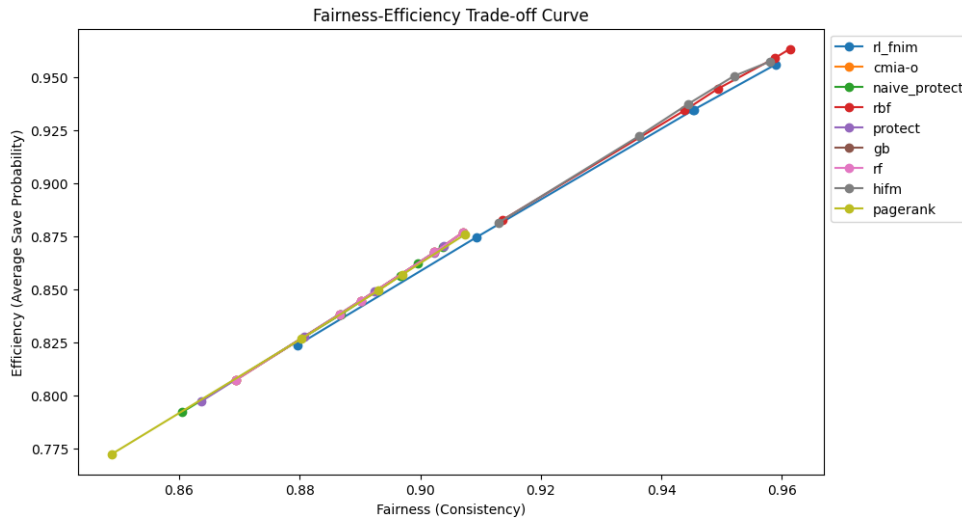
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datasets like SN2. **PageRank** and **RBF** algorithms also perform well, maintaining a strong balance but may not offer the same level of optimization as the newer methods. **CMIA-O** is positioned slightly lower on the curve, indicating it might not be as effective in balancing fairness and efficiency compared to the other algorithms in this dataset.



(c) SN3 dataset

The SN3 dataset exhibits a slightly broader spread of algorithms along the trade-off curve compared to SN1 and SN2. This indicates that there is a bit more variation in how the algorithms perform in balancing fairness and efficiency in this specific network structure. **RL_FNIM** continues to show robust performance, **HIFM** is also positioned near the top of the curve, closely following **RL_FNIM**. It demonstrates strong adaptability and effectiveness in this dataset as well. **CMIA-O** shows slightly lower performance, particularly in fairness, compared to the top-performing algorithms. However, it still maintains a reasonable balance between the two metrics.



(d) SN4 dataset

Fig 5.10 Trade-off analysis of Synthetic datasets

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In the SN4 dataset, the algorithms are tightly clustered along the trade-off curve, similar to previous datasets. The proposed algorithms **RL_FNIM**, **HIFM**, **RF**, and **GB** continue to demonstrate strong performance, effectively balancing fairness and efficiency. **RL_FNIM** and **HIFM** are particularly strong, consistently placing at the top of the trade-off curve. **RF** and **GB** also perform well, indicating their reliability across different datasets. Traditional algorithms like **protect** and **PageRank** show solid performance but may not match the fine-tuned balance provided by the proposed algorithms, particularly in maintaining high fairness while achieving strong efficiency. This consistency across various datasets highlights the robustness of the proposed algorithms, particularly in the complex network structures like SN4.

Overall, In the real-world datasets, the complexity and unpredictability of actual social network structures pose a greater challenge to the algorithms. Despite this, **RL_FNIM** and **HIFM** continued to perform exceptionally well, especially in the Socfb-Caltech36 and CA-AstroPh datasets. These datasets, which reflect more realistic social interactions and network dynamics, showcase the effectiveness of the proposed algorithms in real-world scenarios. **RF** and **GB** also performed competitively, particularly in the CA-AstroPh dataset, which is characterized by its large scale and complex connectivity. Traditional algorithms like **protect** and **PageRank** showed solid performance, particularly in the Email dataset, but they generally did not match the fine-tuned balance of fairness and efficiency demonstrated by the proposed methods.

The synthetic datasets, designed with varying network characteristics and parameters, serve as controlled environments to test the algorithms' performance in balancing fairness and efficiency. Across all four synthetic datasets, **RL_FNIM** and **HIFM** consistently emerged as top performers, maintaining a strong balance between fairness (measured by consistency) and efficiency (measured by average save probability). **RF** and **GB** followed closely behind, demonstrating that these algorithms are highly adaptable to different network topologies. The tight clustering of algorithms along the trade-off curves in datasets like SN2 and SN4 suggests that in more homogeneous networks, the performance differences between the algorithms are minimal. However, in more complex networks like SN3, slight variations in performance become more apparent, with **RL_FNIM** and **HIFM** slightly edging out the others.

In conclusion, the analysis across multiple datasets highlights the superiority of the proposed algorithms, **RL_FNIM**, **HIFM**, **RF**, and **GB** in balancing the trade-offs between fairness and efficiency. **RL_FNIM** and **HIFM** are particularly noteworthy for their consistent top performance across all datasets, proving their capability to handle both synthetic and real-world complexities effectively. The traditional algorithms, while still relevant and effective in certain contexts, do not match the nuanced performance of the newer methods. This study not only underscores the importance of developing algorithms that can adapt to diverse network conditions but also provides a convincing case for the application of **RL_FNIM**, **HIFM**, **RF**, and **GB** in future influence minimization and network analysis tasks.

Chapter 6. Conclusion

6.1 Summary of Dissertation

This dissertation focused on addressing the critical challenge of Negative Influence Minimization (NIM) in social networks, a problem of increasing importance in the digital age where misinformation can spread rapidly and have far-reaching consequences. The primary objective was to develop innovative algorithms that could effectively balance the dual goals of fairness and computational efficiency, thereby ensuring that efforts to mitigate misinformation are both equitable and scalable.

The research began with an extensive review of the existing literature on influence propagation models, influence maximization, and the specific challenges associated with NIM. This foundational work highlighted gaps in current approaches, particularly in how they handle fairness, often favoring well-connected nodes and exacerbating inequalities within the network. Recognizing these limitations, the dissertation proposed a set of four novel algorithms: **Reinforcement Learning for Fair NIM (RL_FNIM)**, **Hybrid Influence and Fairness Maximization (HIFM)**, **Random Forest (RF)**, and **Gradient Boosting (GB)**. Each of these algorithms are designed with the goal of optimizing the trade-off between fairness and efficiency, offering a more balanced approach to influence minimization.

The methodology involved a rigorous process of algorithm development, followed by extensive experimentation using both synthetic and real-world datasets. The synthetic datasets (SN1, SN2, SN3, and SN4) were carefully developed to simulate various network structures and conditions, providing a controlled environment to test the robustness of the proposed algorithms. In contrast, the real-world datasets (Arenas Email, socfb-Caltech36, and CA-AstroPh) presented more complex and unpredictable challenges, reflecting the actual dynamics of social networks.

Through a series of experiments, the dissertation demonstrated that the proposed algorithms consistently outperformed traditional methods in terms of both fairness and efficiency. The trade-off analysis provided deeper insights into how these algorithms manage the balance between fairness and efficiency, highlighting the strengths and weaknesses of each approach.

In addition to the development of new algorithms, the dissertation also contributed to the field by conducting a comprehensive trade-off analysis. This analysis not only confirmed the efficacy of the proposed methods but also provided a framework for future research in understanding the complex dynamics between fairness and efficiency in influence minimization tasks. The results from both the synthetic and real-world datasets underscored the practical applicability of the proposed algorithms, suggesting that they could be effectively deployed in real-world scenarios to combat the spread of misinformation.

Overall, this dissertation advances the state of the art in NIM by offering robust, fairness-aware solutions that are capable of operating efficiently in a variety of network environments. The contributions made through the development of RL_FNIM, HIFM, RF, and GB represent significant progress in the quest to create more equitable and effective strategies for influence minimization in social networks.

6.2 Research Contributions

The key contributions of this dissertation can be summarized as follows:

1. **Development of Four Novel Algorithms:** The dissertation introduced four innovative algorithms RL_FNIM, HIFM, RF, and GB that are developed to address the Negative Influence Minimization

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problem in social networks. These algorithms are tailored to optimize the trade-off between fairness and computational efficiency, making them highly effective in diverse network scenarios.

2. **Comprehensive Trade-off Analysis:** A thorough trade-off analysis was conducted to explore the relationship between fairness and efficiency across different network structures. This analysis not only highlighted the strengths of the proposed algorithms but also provided a deeper understanding of how these two factors interact in the context of influence minimization.
3. **Application to Real-World and Synthetic Datasets:** The proposed algorithms were rigorously tested on both synthetic and real-world datasets, demonstrating their adaptability and effectiveness in varied network environments. The results confirmed that the proposed algorithms could be applied to real-world challenges, offering robust solutions for combating misinformation.
4. **Advancement of Fairness in NIM:** This research advanced the field of fairness in NIM by developing algorithms that ensure equitable treatment of nodes within a network, addressing the limitations of existing methods that often favor well-connected nodes. The proposed algorithms promote an inclusive approach to influence minimization, ensuring that all users are adequately protected from misinformation.

6.3 Future Research and Development

While this dissertation made significant strides in the field of Negative Influence Minimization, there remain several avenues for future research and development:

- **Exploration of Additional Fairness Metrics:** Future research could explore alternative or more nuanced fairness metrics beyond the ones used in this study. This could involve developing metrics that better capture the complexities of fairness in different types of networks or under varying conditions.
- **Integration with Other Machine Learning Techniques:** The proposed algorithms could be enhanced by integrating other advanced machine learning techniques, such as deep learning or advanced reinforcement learning models. This could further improve their adaptability and efficiency in even more complex network scenarios.
- **Real-Time Influence Minimization:** Another promising area for future research is the development of real-time influence minimization algorithms that can adapt dynamically to changes in the network. This would be particularly useful in rapidly evolving social networks where misinformation spreads quickly.
- **Application to Broader Contexts:** While the focus of this dissertation was on social networks, the principles and algorithms developed could be applied to other contexts, such as financial networks, biological networks, or any system where influence propagation plays a critical role.

6.4 Personal Reflections

Embarking on this research journey has been both intellectually stimulating and personally fulfilling. At the outset, I was motivated by a keen interest in the dynamics of influence within social networks and the rising challenge of combating misinformation. However, as I delved deeper into the subject, the importance of balancing fairness with efficiency in algorithm design became increasingly clear. This realization

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fundamentally shaped the direction of my work, leading me to prioritize the development of algorithms that not only perform well but also ensure equitable outcomes for all network participants.

The process of developing the four new algorithms was a challenging yet rewarding experience. Each algorithm presented its own set of technical hurdles, requiring a blend of creativity, rigorous analysis, and persistence. There were moments of difficulty, particularly in fine-tuning the balance between fairness and efficiency. However, overcoming these challenges deepened my understanding of the complexities involved in algorithmic design and strengthened my problem-solving abilities.

Working with both synthetic and real-world datasets provided invaluable insights into the practical application of the proposed algorithms. The real-world datasets, in particular, presented unexpected challenges that highlighted the gap between theoretical models and practical realities. These experiences underscored the importance of adaptability in research, as well as the need for robust algorithms that can perform reliably under diverse and sometimes unpredictable conditions.

As I reflect on the work completed, I am proud of the contributions this dissertation has made to the field of Negative Influence Minimization. However, I also recognize that there is still much to be explored. The potential for further research in this area is vast, and I am excited by the prospect of continuing to contribute to this field in the future. This experience has reinforced my commitment to pursuing research that not only advances knowledge but also has a tangible, positive impact on society.

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APPENDIX A

Algorithmic Implementations of Proposed Methods

Reinforcement Learning for Fair NIM (RL_FNIM):

The Reinforcement Learning for Fair NIM (RL_FNIM) algorithm is designed to optimize the trade-off between fairness and computational efficiency in Negative Influence Minimization tasks. It uses reinforcement learning techniques to iteratively learn the optimal policy for selecting nodes that balance influence reduction with fair distribution across the network (He, K., Qian, L. and Ding, M. (2022)).

Pseudocode:

```

INITIALIZE Q-values for all states and actions
SET learning rate ( $\alpha$ ) and discount factor ( $\gamma$ )
SET exploration rate ( $\epsilon$ ) for epsilon-greedy policy

FOR each episode:
    INITIALIZE the environment and set initial state
    WHILE state is not terminal:
        SELECT action using an epsilon-greedy policy:
            IF random value  $< \epsilon$  THEN
                SELECT a random action
            ELSE
                SELECT action with the highest Q-value for the current state
            END IF
        EXECUTE action, observe reward, and transition to next state
        UPDATE Q-value:
             $Q[\text{state}, \text{action}] = Q[\text{state}, \text{action}] + \alpha * (\text{reward} + \gamma * \text{MAX}(Q[\text{next\_state}, :]) -$ 
             $Q[\text{state}, \text{action}])$ 
        UPDATE state to next_state
    END WHILE
    DECAY  $\epsilon$  to reduce exploration over time
END FOR

```

The RL_FNIM algorithm uses a Q-learning framework to iteratively update the value of taking specific actions (i.e., selecting certain nodes) in different states of the network. The Q-values are adjusted based on the rewards received from the environment, which are linked to both the effectiveness in reducing misinformation and the fairness of the node selection. The epsilon-greedy policy ensures a balance between exploration (trying new strategies) and exploitation (using the best-known strategies), with ϵ decreasing over time to favor exploitation as the algorithm learns.

Hybrid Influence and Fairness Maximization (HIFM):

The Hybrid Influence and Fairness Maximization (HIFM) algorithm aims to select nodes in a social network that maximize both influence spread and fairness. It combines traditional influence maximization techniques with fairness constraints, ensuring that the selected nodes contribute to an equitable distribution of influence across the network.

Pseudocode:

```
INITIALIZE an empty set S for selected nodes
SET the total number of nodes to be selected (k)

FOR each iteration until k nodes are selected:
    CALCULATE composite score for each node:
        Composite score = (Influence Score + Fairness Score) / 2
    SELECT node with the highest composite score
    ADD selected node to set S
    UPDATE the network state based on the newly selected node
    RECALCULATE influence and fairness scores
END FOR

RETURN set S containing selected nodes
```

The HIFM algorithm iteratively selects nodes by evaluating a composite score that equally weighs influence potential and fairness. The algorithm ensures that the selected nodes not only reduce misinformation effectively but also distribute influence fairly across the network, preventing biases towards well-connected nodes. The selection process continues until the specified number of nodes (k) is chosen, with scores being recalculated after each selection to reflect the updated network state.

Random Forest (RF):

The Random Forest algorithm applies ensemble learning techniques to predict node influence in a social network. By constructing multiple decision trees, the algorithm estimates the influence potential of each node while incorporating fairness considerations into the final decision-making process.

Pseudocode:

```
PREPROCESS network data:
    Extract features relevant to node influence and fairness

SPLIT data into training and testing sets

TRAIN Random Forest model:
    FOR each tree in the forest:
        SELECT a random subset of features
        GROW the tree using the training data:
            FOR each node in the tree:
                SELECT the best split based on information gain
            END FOR
        END FOR

PREDICT node influence:
    USE the trained forest to predict influence scores for each node

INTEGRATE fairness:
    ADJUST the predicted influence scores to account for fairness metrics:
        Fairness-adjusted score = (Predicted Influence Score + Fairness Score) / 2

SELECT top-k nodes based on fairness-adjusted scores

RETURN set of selected nodes
```

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The **RF** algorithm leverages the power of ensemble learning by constructing multiple decision trees, each trained on a random subset of features. These trees collectively predict the influence of each node, with the final decision being an average of these predictions. Fairness is integrated into the process by adjusting the predicted influence scores based on fairness metrics, ensuring that the selection of nodes is not biased towards highly connected nodes, but instead reflects a balanced and fair distribution across the network.

Gradient Boosting (GB):

The Gradient Boosting (GB) algorithm is a machine learning technique that builds models sequentially, each correcting the errors of its predecessor. In the context of NIM, this algorithm focuses on minimizing the residual errors in influence prediction while integrating fairness into the final model predictions.

Pseudocode:

```
PREPROCESS network data:
    Extract features relevant to node influence and fairness

INITIALIZE predictions to 0 for all nodes

FOR each boosting iteration:
    CALCULATE residuals based on current predictions:
        Residual = True Influence Score - Current Prediction
    TRAIN a weak learner (decision tree) on residuals:
        FOR each node in the tree:
            SELECT the best split based on minimizing the residuals
        END FOR
    UPDATE predictions by adding the weighted predictions of the weak learner:
        Prediction = Previous Prediction + Learning Rate * Tree Prediction
    END FOR

PREDICT node influence using the final boosted model

INTEGRATE fairness:
    ADJUST the predicted influence scores to account for fairness metrics:
        Fairness-adjusted score = (Predicted Influence Score + Fairness Score) / 2

SELECT top-k nodes based on fairness-adjusted scores

RETURN set of selected nodes
```

Gradient Boosting (GB) works by iteratively improving predictions made by weak learners (small decision trees). Each iteration focuses on the residuals (errors) from previous predictions, gradually improving the model's accuracy. The final model combines these corrections to make more accurate predictions of node influence. Fairness adjustments are made to these predictions to ensure that the selected nodes reflect a fair distribution across the network, balancing influence reduction with equitable node selection.